



# Interpreting the Distributional Successor Measure as a model for the Hippocampus

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**Code:** The code can be found on GitHub at [dsm-hippocampus](https://github.com/dsm-hippocampus)

## **Abstract**

Navigation is a key cognitive function thought to be supported by hippocampal place cells, which serve as a predictive map for spatial environments. Traditional models, such as the Successor Representation (SR), have provided valuable insights into how the hippocampus encodes these environments, but face limitations in capturing the dynamic and flexible nature of place cells, particularly their ability to adapt to changes and encode distributions over future states. To address these limitations, this thesis explores the Distributional Successor Measure (DSM), an extension of the SR framework that models distributions over possible future states, offering a closer alignment with biological neural systems. This work employs neuroscientific analysis techniques to examine the DSM's internal representations with respect to environmental features and investigates whether the DSM can mirror place cell activity while evaluating its robustness to environmental changes.

Results demonstrate that the DSM provides richer and more flexible predictive representations of place cells and exhibits greater resilience to environmental changes compared to the SR. These findings suggest that the DSM more closely aligns with biological observations of hippocampal function and provides a robust model for how the hippocampus encodes predictive maps in a distributional manner. This thesis contributes to a deeper understanding of the relationship between artificial intelligence models and biological neural systems, highlighting the potential of the DSM to bridge this gap.

**Code:** The code can be found on GitHub at [dsm-hippocampus](https://github.com/dsm-hippocampus)

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# Chapter 1

## Introduction

Understanding how the brain represents and processes spatial information to navigate the world has been a central focus in neuroscience. Animals, particularly mammals, possess a remarkable ability to navigate and adapt to their surroundings, ensuring survival by finding sustenance, avoiding predators, and returning to shelter. This often involves complex behaviors such as taking detours or shortcuts when faced with novel challenges. Fundamentally, these abilities are thought to rely on the brain's capacity to construct flexible internal representations of space—what are commonly referred to as cognitive maps.

Edward Tolman first introduced the concept of cognitive maps in 1948 [1], theorizing that animals navigate using internal representations of their environments rather than simply reacting through reflexive behaviors. His early model conceptualized these maps as static, Euclidean spaces that animals use to chart out their environment. This theory gained substantial support with the discovery of place cells in the hippocampus by O'Keefe and Dostrovsky in 1971 [2]. Place cells, which fire in response to specific spatial locations, provided compelling evidence that the hippocampus functions as a spatial map underpinning navigation.

Subsequent research revealed additional neural components contributing to spatial representation, such as head direction cells, grid cells and boundary cells. These discoveries expanded our understanding of the neural circuitry involved in processing spatial information, particularly the hippocampal-entorhinal system. Notably, this spatial circuitry is conserved across species, including rodents and primates, underscoring the fundamental role of spatial representation in mammalian brains. The organized manner

in which these cells encode space enables animals to navigate both familiar and novel environments, supporting critical behaviors such as decision-making and planning.

## Motivation

While these discoveries have significantly advanced our understanding of spatial representation, they also reveal that traditional models viewing cognitive maps as static, Euclidean spaces may be overly simplistic. The reality of spatial navigation in animals suggests a more dynamic and flexible system. Experimental studies, such as those identifying time cells [3], splitter cells [4] and event-specific representations [5], demonstrate that hippocampal cells can encode more than just current spatial information—they incorporate potential future states and task-related information. Such flexibility implies that the brain employs predictive models, extending beyond static representations to dynamically encode possible future experiences. This shift in perspective aligns with computational models in reinforcement learning (RL), particularly the Successor Representation (SR) [6], which has been proposed to explain this predictive nature [7], by encoding expected future states based on an agent’s current state.

The SR model provides a computational framework for understanding predictive spatial representations in the brain, and has been shown to capture many important properties of place cells. However, despite its utility, the classic SR model has inherent limitations in its ability to generalize across novel environments and adapt to unexpected changes in spatial layout. Specifically, the SR can use its cached representations to readily solve reward revaluation, but struggles with transition or policy revaluation [8], failing to fully account for complex behaviors observed in place cells, such as partial remapping with environmental changes.

The SR model continues to be refined and extended to better match neurobiological observations. Recent work has explored how SR can be implemented in recurrent neural networks, potentially bridging gaps between SR and observed place cell properties. It has also been found that the SR can be encoded in the brain through a few biologically plausible mechanisms [9–11]. While SR is valuable for understanding certain aspects of spatial cognition and memory, it should be considered part of a broader framework that includes additional computational principles to fully capture the complexity of place cells and related neural systems.

The Distributional Successor Measure (DSM) [12] is a recent framework designed to overcome some of the limitations of Distributional Reinforcement Learning (DRL) models by enabling zero-shot policy evaluation—allowing policies to be evaluated for new rewards without retraining. DSM extends the SR concept to DRL by decoupling reward from transition dynamics, enabling evaluation of the entire distribution of returns under different conditions and improving risk-aware evaluations across various tasks.

Unlike traditional SR models that rely on a single predictive map, DSM distributes predictions across multiple components or “atoms”, each capturing different aspects of the environment’s dynamics. This distributional approach allows DSM to represent the entire distribution of possible future states, not just the expected future state. By leveraging multiple components, the DSM should hypothetically handle uncertainty better and adapt more efficiently to changes in the environment—properties that mirror the hippocampus’s capacity for flexible and adaptive navigation.

Integrating multiple predictive distributions allows the DSM to capture the variability and complexity of future states in a manner that closely aligns with the distributed, heterogeneous nature of neural representations in the hippocampus. This richer encoding of space may explain how the hippocampus encodes various environmental features, including non-linearities and contextual dependencies in real-world navigation. Moreover, DSM’s ability to encode multiple future possibilities simultaneously also reflects the hippocampus’s predictive role, representing potential future paths based on environmental structure and the animal’s goals. This distributed nature could provide insight into phenomena like place cell remapping and context-dependent spatial coding, aspects that are difficult to capture with SR models alone.

### **Current distributional theories in the brain:**

Recent neuroscience and psychology research suggests that reward-based decisions are informed by more than just simple, incremental updates associated with individual actions; they rely on flexible memory mechanisms capable of integrating past experiences over multiple timescales [13]. Neural signals, previously thought to map directly to reward prediction errors, exhibit diversity that cannot be fully explained by traditional RL models. Instead, this variability is better accounted for by distributional theories [14,15], which enable mechanisms integrating diverse sources of information beyond simple summary statistics.

Building on these insights, we propose that the Distributional Successor Model (DSM) may incorporate latent variables that better represent the history of states and actions. This could reflect how past locations are stored in memory and influence future choices, aligning with the idea that internal mechanisms in the brain do more than store a single expected future state. Such a structure would parallel the separation of reward-guided decisions and action-history processing symbolic of traditional cognitive models [13].

One of the challenges in applying artificial neural networks (ANNs), including models like DSM, to modeling biological systems is interpretability. Classic cognitive models have parameters with clear roles, but they often perform poorly in areas where ANNs excel at learning complex patterns. However, the internal computations of ANNs can be harder to parse. Here, we use neuroscientific techniques to understand the representations in the DSM model to gain insights into how the brain represents trajectories and states in an environment and how these representations may influence longer-term strategies for navigation.

## Aims and research directions

Building on the foundations of cognitive maps and the SR framework, the DSM model offers a significant advancement in modeling hippocampal predictive representations. This thesis aims to explore this potential of DSM by integrating the distributional framework with place cell data.

The investigation has two key aims.

**Analyzing DSM Representations in Controlled Environments:** We plan to use traditional analysis tools in neuroscience to understand the DSM model’s mechanisms, similar to how neuroscientists record neuronal activity while measuring behavior in response to different stimuli. By employing the inverted pendulum task—a well-understood benchmark in reinforcement learning—we can analyze the DSM’s representations in a controlled setting. This approach will help us interpret what the learned DSM model’s neurons may represent concerning the pendulum’s state space, building intuition about the model’s workings before progressing to more complex spatial environments.

**Modeling Place Cell Dynamics with DSM:** We aim to explore whether the DSM

model can effectively simulate place cell dynamics and remapping behaviors observed in hippocampal neurons. In the context of spatial navigation, we simulate scenarios involving foraging behavior in rodents by generating realistic locomotion patterns and synthetic place cell data in complex continuous environments.

Drawing on research that suggests that the brain encodes feature-specific prediction errors in the form of heterogeneous dopaminergic responses [15], we speculate that in DSM’s structure, different atoms may correlate with specific features of the environments. By analyzing the activations of different atoms, we aim to determine whether they are oriented toward predicting various environmental features or behavioral patterns

The experiments investigate the DSM’s ability to learn different predictive maps from the place field activity at discretized points in the environment. By simulating various movement policies in similar environmental conditions, we will compare DSM’s performance with that of the conventional SR model, assessing its ability to generalize across environments in the form of changing cues by changing place cell centers. Furthermore, we will investigate whether DSM’s hidden layers reveal interpretable patterns of neural activity that align with known hippocampal features. Studying the layers before the hippocampal output can provide insights into its functionality, potentially indicating how it represents various dynamics in its observation space. Understanding these representations will help us understand biological intelligence particularly as theories emerge in how dopamine and serotonin act as distributed RPEs.

Through these simulations, the study aims to determine whether DSM offers a closer approximation to how the brain represents and processes spatial information, not only predicting single future states but an entire range of possibilities.

In conclusion, this thesis seeks to explore the distributed successor measure (DSM) model from a neuroscientific perspective, focusing on its potential to model the distributional nature of hippocampal place cells. By leveraging the distributional framework, we aim to develop a computational model that captures the brain’s capacity to predict all possibilities of future states with respect to past experiences. We aim to determine whether DSM can shed light on how the hippocampus encodes predictive distributions over future trajectories for any state in its environment, providing a richer understand-

ing of how past experiences (learnt policy) shapes future behavior. We will use traditional neuroscientific tools to analyze the DSM model's internal representations. By examining DSM's representations in a simple environment like the inverted pendulum task, we hope to have basic measures of interpreting the model and identify their utility in modeling brain function. The experiments investigate the DSM's ability to learn different predictive maps from the place field activity at discretized points in the environment. Further we explore how the remapping of space through shifting place cell centres may affect the distributional encoding.

## Chapter 2

# Background and Literature Review

This chapter provides an overview of the study background and literature, beginning with an exploration of the neuroscience of the hippocampus and its cell types, particularly focusing on place cells and its predictive properties. We discuss several models that offer a framework for modeling hippocampal functions, before exploring RL models.

The chapter then covers key RL concepts—model-free and model-based learning—before focusing on the Successor Representation (SR), which serves as an intermediate framework combining elements of both. We examine how SR aligns with the predictive function of place cells in the hippocampus and explore the limitations of SR in capturing the full flexibility of place cell behavior. To address these limitations, we introduce more recent advances, such as gamma models and distributional reinforcement learning (DRL), which extend SR by providing continuous and distributional perspectives. While gamma models offer a continuous extension of SR, DRL introduces a way to model the variability in future outcomes. This thesis introduces the Distributional Successor Measure (DSM) as a novel framework for modeling hippocampal place cells. By integrating the distributional approach of RL with gamma models, DSM captures the range of possible future states and offers a richer, more flexible prediction of place cell activity, addressing the limitations of SR and potentially offering insights into the distributed predictive nature of the hippocampus.

## 2.1 Neurophysiology: The Hippocampus and Place Cells

### 2.1.1 The Hippocampus and Its Role in Spatial Navigation

Understanding spatial navigation requires an in-depth exploration of the hippocampus, a brain region central to cognitive neuroscience research since the mid-20th century [16]. The hippocampus, a seahorse-shaped structure located in the brain's medial temporal lobe, plays a critical role in spatial navigation and episodic memory. This was first evidenced by classic studies on amnesic patients like H.M., whose hippocampal lesions led to profound deficits in forming spatial and episodic memories [17].

The hippocampus, together with the entorhinal cortex, forms the hippocampal-entorhinal system, integral to encoding and processing spatial and temporal information in mammals. This system is characterized by intricate loops of connectivity that integrate sensory information from multiple sources. A key component is the tri-synaptic loop, (illustrated in Figure 2.1), a neural circuit involving the dentate gyrus (DG), CA3, and CA1 regions of the hippocampus [18]. In this circuit, the entorhinal cortex serves as the primary gateway for spatial and sensory information entering the hippocampus. Inputs from the medial entorhinal cortex (MEC) and lateral entorhinal cortex (LEC) flow first to the DG, then progress to CA3, and finally reach CA1 before projecting back to the entorhinal cortex and other cortical areas [19]. This circuit facilitates communication between the hippocampus and cortical regions and supports the encoding of spatial information essential for forming cognitive maps.

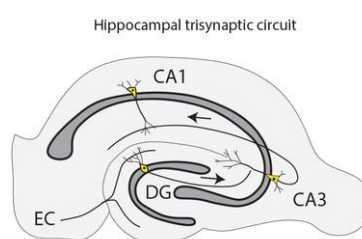


Figure 2.1: The tri-synaptic loop in the hippocampus. *Figure adapted from [20]*

Multiple lesion studies have demonstrated the crucial role of the hippocampus in navigation [21,22]. Animals with hippocampal lesions struggle with place-based navigation strategies, which rely on the spatial relationships between environmental cues to form internal representations—or cognitive maps—of their environment [23]. In-

stead, these animals resort to response-based navigation strategies, relying on habitual motor actions such as consistently turning left or right at certain junctions, regardless of external cues. In contrast, animals with intact hippocampal circuits are assumed to use cognitive maps to navigate toward remembered goals based on the spatial relationships between environmental cues. This suggests that the hippocampus is essential for place-based navigation and that when it is compromised, animals default to simpler, habit-based strategies that do not require spatial representations.

### **2.1.2 Specialized Neurons in Spatial Navigation**

Exploration into the hippocampal-entorhinal network has revealed the existence of several specialized neurons [24]:

- **Place Cells:** Located primarily in the CA1 and CA3 regions of the hippocampus, place cells fire when an animal occupies a specific location in its environment [2]. They are central to the formation of cognitive maps and are the focus of this project.
- **Grid Cells:** Found in the MEC, grid cells fire at multiple locations that form a hexagonal lattice across the environment, providing a metric for spatial navigation [25].
- **Head Direction Cells:** These cells encode the animal's directional orientation, functioning like an internal compass [26]
- **Boundary vector cells:** These cells respond to the animal's proximity to environmental boundaries [27]

Together, these neurons form a comprehensive spatial navigation system, integrating various types of information to support navigation.

### **2.1.3 Place Cells: Discovery and Properties**

Given that this project involves modeling place cells, a detailed examination of their properties and functions is essential.

The identification of place cells by O'Keefe and Dostrovsky in 1971 [2] was a pivotal discovery in understanding spatial navigation within the hippocampal formation.

These pyramidal neurons fire selectively when an animal occupies a specific location in its environment, creating what is known as a “place field”. This firing pattern provides compelling evidence that the hippocampus functions as a cognitive map, aligning with Edward Tolman’s earlier concept of mental representations guiding behavior [1].

In typical experiments, the activity of place cells in rats is recorded while they engage in navigation tasks or forage in a large open arena where food pellets are randomly dispersed to encourage thorough exploration of the environment [2]. The firing of a place cell is spatially localized, as shown in Figure 2.2, where action potentials occur predominantly in a specific area, forming a place field. The spatial representation of place cells is largely independent of the animal’s orientation, making it “allocentric” (external-referenced). This allows animals to navigate using internal maps based on external objects or landmarks.

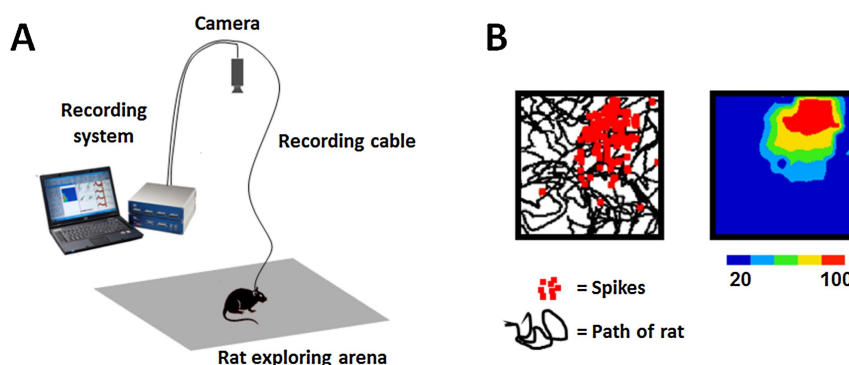


Figure 2.2: (A) Typical experiment recording single neurons from freely exploring rats with implanted electrodes. (B) In one trial, the left plot shows the rat’s path, with action potentials (red squares) mostly occurring in the northeast. The right plot depicts this data as a heatmap of the firing rate, adjusted for dwell time, highlighting a spatially localized firing area, known as a place field. *Figure adapted from [28]*

Place cells are not merely responsive to isolated cues but integrate information from various sources, including distal landmarks, local boundaries, objects, odors, and self-motion cues like vestibular input [18]. By combining diverse cues, place cells seem to construct a coherent spatial map that supports flexible learning and cognitive processes. Research has shown that place cells are not limited to rats; similar cells have been discovered in other mammal species, such as mice, bats, monkeys, and humans, indicating the broad applicability of this spatial coding mechanism across species [29–32].

## Variability and Flexibility in Place Cell Activity

Place cells are not a homogenous population; their functional properties vary depending on their anatomical location within the hippocampus and the specific environmental context. For example, place cells in the ventral hippocampus tend to have larger, less stable fields than those in the dorsal hippocampus, reflecting the distinct roles of these regions [33]. Models replicating this behavior [34] adjust place field sizes dynamically based on environmental complexity, activating larger fields in open spaces and smaller fields near obstacles or goals.

A feature of place cell activity is overdispersion, where their firing patterns vary even during consistent trajectories in stable fixed environments, indicating the integration of additional cognitive processes. In larger environments, place cells can develop multiple, unevenly spaced fields, adapting their coding properties to the scale and complexity of the space. Although place fields typically do not exhibit directionality, they can develop directional firing patterns with experience in linear environments [18].

A key property of place cells is their ability to “remap”, changing their firing patterns in response to alterations in the environment or context [35]. This process enables the hippocampus to generate multiple distinct representations of the same space, serving as a pattern separation mechanism crucial for discriminating between different environments or encoding specific temporal or behavioral variables within a single environment [36,37]. Place cells also exhibit a “global remapping” where the positions of the place fields change [38]. This flexibility of hippocampal representations underscores its role in navigation and memory processing. This flexibility is evident from experiments where changes to the geometry of an environment, such as the elongation of walls, resulted in place fields expanding or adjusting accordingly [39,40]. Similarly, the introduction of an aversive stimulus led to significant shifts in place fields or the emergence of new fields, indicating that the hippocampal representation of space is influenced by motivational valence and associated behavioral responses [41].

Moreover, place cell research encompasses several coding concepts, such as dedicated coding, where individual neurons independently signal specific stimuli or concepts; ensemble coding, where groups of neurons jointly encode information; and manifold coding, which suggests that neural population activity forms stable geometric patterns in high-dimensional space across different conditions. [42]

## Theta rhythms and prospective coding in the hippocampus

As an animal moves through space, the hippocampus faces the computational challenge of arranging the discrete activations of place cells into temporal sequences to represent trajectories. The hippocampus generates oscillations in the theta range (4–8 Hz) during active behaviors like running and exploration, providing a temporal framework that links spatial locations into a coherent, predictive map [43].

During theta oscillations, phenomena such as theta phase precession occur, integrating information about both the timing and location of an animal's movement. Theta phase precession is used to describe when as an animal moves through a place field, individual place cells fire at progressively earlier phases of the theta cycle, with peak firing of each place cell (when animal is at the centre of its place field) occurring in the middle of the cycle. This phase coordination ensures that place cells encode trajectories as ordered sequences, aiding the brain's ability to plan future actions based on past and current experiences. At critical decision points, these theta rhythms facilitate forward sweeps of place cell activity, allowing the hippocampus to momentarily represent potential future locations before the animal reaches them, effectively providing a predictive framework for decision-making [44]

Studies have also suggested that sharp wave-ripple (SWR) events, which occur during offline periods such as rest or sleep, allow for the replay of hippocampal activity sequences. [45] These are thought to support sequential planning by simulating potential future trajectories (preplay [46]) or recalling past experiences, facilitating the formation of spatial and episodic memories. During active navigation, these same sequences assist with forward planning, where the brain uses theta rhythms to internally simulate upcoming states and guide decision-making.

Building upon this predictive capacity, place cells can thus exhibit prospective or retrospective coding, altering their firing based on anticipated future events or recent experiences [47]. For a concrete example, on a T-maze task, some place cells ("prospective splitter cells") fire differently depending on whether the animal plans to turn left or right at an upcoming choice point, even though the current location is the same [48]. These findings suggest that place cells process broader information beyond just euclidean maps, involving elements of memory and prediction [3,49].

### 2.1.4 Critical Analysis of Hippocampal Function and Modeling

In summary, the hippocampus, through its intricate network of specialized cells, plays a pivotal role in spatial navigation and memory. Its capacity for dynamic spatial representation and predictive coding via place cells offers profound insights into how biological systems anticipate and adapt to their environments. Despite significant advances in understanding hippocampal place cells, modeling their features and flexibility remains challenging.

Over the decades, place cells have been the subject of various modeling approaches. Early theories, such as O’Keefe and Nadel’s Cognitive Map Theory (1978) [50], proposed that place cells form the basis of a spatial memory system in the hippocampus, emphasizing its role in allocentric mapping, where locations are represented independently of the observer’s position during exploration. While foundational, this theory did not fully account for the dynamic and adaptable nature of place cell representations observed in experiments.

Subsequent mechanistic models emerged to explain different aspects of place cell function. Continuous Attractor Neural Network (CANN) models demonstrated how stable spatial representations could arise from network dynamics, proposing that place cells form stable activity patterns within an attractor map [51]. These models suggest that the collective activity of place cells can maintain a representation of position through recurrent connectivity. Other models emphasized the role of self-motion cues in updating spatial representations [52].

While these traditional models provided valuable insights into various aspects of place cell function, they struggle to account for the various features like remapping and prospective coding. This limitation highlighted the need for more sophisticated modeling approaches that could capture the complex, adaptive nature of hippocampal function.

Reinforcement learning (RL) offers a computational framework that mirrors hippocampal functions by predicting future rewards and guiding decision-making based on past experiences. The parallels between these two domains have inspired a number of computational models attempting to bridge neuroscience and artificial intelligence. Models like the Successor Representation (SR) provide direct analogies to the predictive functions of place cells by encoding expected future state occupancy.

In the following section, we explore the foundational concepts of RL and its applications in modeling the predictive, distributed nature of hippocampal function. We begin with model-based and model-free learning, and then discuss more advanced models that aim to capture the complexity of hippocampal behavior.

## 2.2 Reinforcement Learning

Reinforcement Learning (RL) is a computational framework for modeling how agents learn to make decisions by interacting with their environment to maximize cumulative future rewards, known as return. This problem manifests across various domains—from playing chess, where the goal is to maximize the probability of winning the game, to navigation tasks where the state represents a location in the environment and the objective is to maximize the accumulation of rewards.

The roots of RL can be traced back to early 20th-century debates in animal learning psychology [53]. Pavlov’s work on classical conditioning established the principles of associative learning. Thorndike’s Law of Effect and Skinner’s Operant Conditioning emphasized the role of consequences in shaping behavior, aligning closely with RL’s reward-based learning. In contrast, Tolman(1948) [1] proposed that actions are planned using internal representations of environmental contingencies, or “cognitive maps”.

Modern cognitive neuroscience reconciles these perspectives, supporting a dual-process theory of learning and decision-making that parallels a crucial classification in RL algorithms: “model-free” (habitual) and “model-based” (goal-directed) approaches. Model-free algorithms learn a direct mapping from perceptual inputs to action outputs based on experienced rewards, without explicit knowledge of the environment’s dynamics. In contrast, model-based algorithms learn a model of the environment’s transition dynamics and reward structure, using this information to plan actions by forecasting their outcomes [54]. This dichotomy has significantly influenced neuroscience research, with different brain regions being associated with these two types of learning.

### 2.2.1 Fundamentals of Reinforcement Learning

The key components of RL include the agent, the environment, states, actions, rewards, and policies. At each time step  $t$ , the agent observes a state  $s_t$ , selects an action  $a_t$ , receives a reward  $r_t$ , and transitions to a new state  $s_{t+1}$ . The agent's objective is to learn a policy  $\pi(a | s)$  which maps states  $s$  to actions  $a$ , aiming to maximize the expected cumulative future rewards. This can be learned through different approaches. The RL problem is typically formalized using a Markov Decision Process (MDP), defined by the tuple  $\langle S, A, P, R, \gamma \rangle$  where:

- $S$ : Set of possible states
- $A$ : Set of possible actions
- $P(s' | s, a)$ : Transition probability function; the probability of transitioning to state  $s'$  from state  $s$  after action  $a$ .
- $R(s, a)$ : Reward function; the expected reward received after taking action  $a$  in state  $s$ .
- $\gamma \in [0, 1]$ : Discount factor that balances immediate and future rewards

The goal is to find an optimal policy  $\pi^*$  that specifies the best action to take in each state to maximize expected return. A value function quantifies the expected return from a state or state-action pair under a particular policy. The value  $V(s)$  of each state  $s$  in an environment is represented as the expected sum of future rewards starting from that state as described in Eq.(2.1).

$$V(s) = \mathbb{E} \left[ \sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right] \quad (2.1)$$

These methods are often classified into model-free and model-based RL, with the successor representation (SR) offering a hybrid model that blends features of both.

### 2.2.2 Model-Free and Model-Based Reinforcement Learning

Model-free RL methods are driven by experience, learning optimal policies or value functions without explicitly modeling the environment's dynamics. These methods are

relatively simple and can be effective in large or complex environments where it is challenging to model the environment explicitly. However, they often require extensive exploration and can be slow to converge. Additionally, they lack flexibility when environmental dynamics change, often resulting in habitual behavior.

One common model-free method is Temporal Difference (TD) Learning, directly updates the value function based on the observed reward and the estimated value of the next state, using sampled experience. The expectation is approximated through repeated interactions with the environment. The value function  $V(s)$  is updated iteratively using the Bellman equation:

$$V_{t+1}(s_t) \leftarrow V_t(s_t) + \alpha [r_t + \gamma V_t(s_{t+1}) - V_t(s_t)]$$

where  $\alpha$  is the learning rate,  $r_t$  is the immediate reward and  $V_t(s_{t+1})$  is the estimated value of the next state.

This mechanism parallels the biological reward prediction errors observed in dopaminergic neurons, particularly in the Ventral Tegmental Area (VTA), which are thought to encode such TD errors [55].

In Model-Based RL, the agent builds an internal model of the environment, learning both the transition dynamics  $P(s' | s, a)$  and the reward function  $R(s, a)$ . Using this model, the agent can plan by simulating future trajectories, calculating the value function by explicitly computing the expectation  $\mathbb{E}$  over all possible future states based on the learned or given transition model. In a biological context, this process is thought to align with how the hippocampus might simulate possible future states during sharp wave ripple events, a phenomenon observed during rest or sleep, which is believed to support planning and memory consolidation [56].

### 2.2.3 The Successor Representation

The Successor Representation (SR), introduced by Peter Dayan in 1993 [6], provides a middle ground between model-free and model-based learning by decomposing the value function into the long-run transition dynamics and corresponding reward statistics of the environment. The transition statistics are captured by the matrix  $M(s, s')$  representing the expected discounted number of future visits to each state  $s'$  when starting

from state  $s$ , under a given policy.

The SR decouples the environment's dynamics from the reward function, making it more adaptable to dynamic environments where rewards may change but the transition structure remains stable. The value function is computed as

$$V(s) = \sum_{s'} M(s, s') R(s')$$

Here,  $M(s, s')$  represents the successor representation and  $R(s')$  is the reward function that assigns value to each state. The SR is updated using a temporal difference (TD) learning rule,, similar to model-free methods, but maintains a predictive model of future state occupancy. The TD update for the SR can be expressed as:

$$M(s_t, s') \leftarrow M(s_t, s') + \alpha (1(s_{t+1} = s') + \gamma M(s_{t+1}, s') - M(s_t, s'))$$

This can also be formulated as

$$M(s_t, a_t) \leftarrow \mathbb{E}_{s_{t+1}} [\mathbf{1}(s_{t+1}) + \gamma M(s_{t+1})]$$

Here, the learning rate is controlled by  $\alpha$ . The time horizon for future states is governed by the discount factor  $\gamma$  which controls the importance given to distant future states relative to immediate ones, capturing both long-term and short-term dynamics.

### **Link to Hippocampal Place Cells and Neural Evidence**

Research has drawn strong parallels between the SR and the predictive role of the hippocampus, particularly in spatial navigation. Studies have proposed that the hippocampus encodes a Successor Representation of space, wherein long-term predictions of state transitions are represented across spatial environments [7]. By storing long-run transition probabilities between states, SR models provide a computational explanation for how the brain might anticipate future locations based on current and past experiences.

In spatial navigation tasks, this is analogous to predicting the sequence of positions an animal will traverse as it moves through its environment. This predictive mapping is not merely theoretical; it has been linked to experimental evidence, with studies showing that the firing patterns of place cells are modulated by factors such as the animal's

movement policy and reward location.

One example is an experimental study which shows the phenomenon of backward expansion of place cells [57], where the place fields shift to include locations the animal recently traversed, especially as the animal consistently moves in a particular direction. This backward skewing implies that place cells represent not only the animal’s present location but also integrate recent past locations into their firing patterns. This aligns with the SR’s predictive nature, where place cells map future states based on the agent’s expected trajectory.

The sensory prediction error (SPE) hypothesis [58] provides additional support for the relevance of SR in neural computation: this hypothesis suggests that dopamine (DA) neurons play a central role in encoding not only reward prediction errors but also errors in sensory predictions, thereby updating the agent’s expectations about future states. In the SR framework, these prediction errors allow for the continuous updating of the successor matrix, thereby enabling the hippocampus to adjust its spatial predictions in response to changing environments or task demands.

Despite these insights into the predictive nature of hippocampal representations, the typical SR formulation has significant limitations. In its basic form, the SR discretizes space into discrete states, with each state corresponding to a specific location within the environment. The agent learns the SR by transitioning through these states, predicting the future occupancy of each location under a given policy, and storing this information in a tabular format. However, this approach faces challenges in real-world scenarios. In large-scale or dynamically changing environments, a fixed tabular SR struggles to capture rapid transitions or represent irregular state spaces effectively. This limitation becomes particularly evident in situations where dimensional distortions occur, significantly altering hippocampal representations [59].

Moreover, the discrete nature of this formulation doesn’t align well with the continuous, fluid nature of spatial representations observed in the brain. To address this, using feature-based representations of state space, rather than direct discretization, allows the SR to generalize across similar states, overcoming the issue of dimensionality.

One such model is the BVC-SR model [59], which created a continuous SR version by learning a SR from a basis set of known neurobiological features—boundary vector cells

(BVCs). This approach describes place cell firing as the successor features of the SR, with grid cells forming a low-dimensional representation of these successor features. While this model overcomes some limitations of its predecessor in capturing biological features under different environmental manipulations, it still relies on a predefined set of features and doesn't fully address the challenge of generating future states in complex, continuous environments.

To further bridge the gap between discrete SR formulations and the continuous nature of real-world environments, researchers have developed more sophisticated models that can generate possible future states in high-dimensional, continuous state spaces.

### **Gamma models**

The  $\gamma$ -model [60] was introduced as a continuous, generative version of the SR, implemented using generative modeling techniques like generative adversarial networks (GANs) or normalizing flows to handle high dimensional continuous state spaces. It is trained using a generative version of temporal difference learning and allows for sampling from this distribution to generate possible future states directly.

In gamma models, the discounted occupancy (or successor occupancy) represents the expected number of future visits to a state under a given policy and is expressed as a weighted mixture over an infinite series of multi-step models. Setting  $\gamma$  to zero recovers the standard single-step model, while any  $\gamma$  within the interval  $(0, 1)$  induces a model with an infinite, probabilistic horizon. As  $\gamma$  increases, the model's predictive lookahead qualitatively extends further into the future. Value estimation in  $\gamma$ -models is performed in a single forward pass, similar to successor features and deep SR, with value expressed as an inner product.

While  $\gamma$  models offer improvements as a continuous variant of the SR, they still lack the flexibility that model-based methods offer. Its policy dependence and failure to adapt to transition and policy revaluation, presents a challenge for understanding the full scope of hippocampal function. It fails to capture many place cells behaviours such as remapping and dynamic coding of environmental changes. To address these challenges and further bridge our understanding of hippocampal function, researchers have turned to other RL concepts, such as Distributional Reinforcement Learning.

## 2.2.4 Distributional Reinforcement Learning

Traditional Reinforcement Learning (RL) algorithms estimate the expected return  $G^\pi(x)$ , defined as the cumulative discounted sum of future rewards starting from state  $x$  under policy  $\pi$ . This return is often expressed as a single scalar value:

$$G^\pi(x) = \sum_{t=0}^{\infty} \gamma^t R_t \quad \text{where} \quad X_0 = x$$

where  $R_t$  represents the reward at time step  $t$ , and  $\gamma \in [0, 1]$  is the discount factor

However, collapsing future rewards into a single scalar value overlooks crucial information about the variability and uncertainty in outcomes. Modeling the full distribution of returns becomes particularly important in scenarios involving function approximation or inherent stochasticity.

**Distributional Reinforcement Learning (DRL)** [61] addresses these challenges by modeling the entire distribution of possible future returns  $G^\pi(x)$ . The distributional Bellman equation extends the traditional Bellman equation to represent the full return distribution, where the return is modeled as a random variable:

$$G^\pi(x) \stackrel{D}{=} R + \gamma G^\pi(X')$$

where  $G^\pi(X')$  is the distribution of returns starting from the next state  $X'$ , and  $\stackrel{D}{=}$  denotes equality in distribution.

Due to the computational infeasibility of storing infinite-dimensional probability distributions, the distribution of returns is approximated using categorical representations or quantile approximations. The return distribution are discretized into a finite set of values or “atoms”, where each atom corresponds to a specific value within a bounded range. In the C51 algorithm [61], the return distribution is approximated using 51 discrete atoms, and the probability associated with each atom is learned using a softmax function to ensure the total probability sums to one.

Distributional RL provides several advantages:

- **Enhanced Learning Signals:** By modeling the entire distribution, agents receive a rich set of auxiliary predictions, such as the probability of specific returns, which

can improve the sample efficiency and generalization capabilities of the agent [62].

- **Handling Uncertainty:** Even in deterministic settings, state aliasing or partial observability introduces effective stochasticity. Recent works suggest that partial observability may even be implicit in goal-driven tasks [20]. Explicitly modeling variability and uncertainty allows agents to make more robust decisions in stochastic environments by enabling better focus on task-relevant aspects.
- **Biological Plausibility:** Evidence suggests that dopaminergic neurons exhibit heterogeneity in their responses to rewards, aligning with distributional RL models [14]

In neural systems, complex and interconnected representations of the environment are often processed in parallel, leading to robust spatial and temporal representations of the external world. For instance, dopaminergic heterogeneity may manifest at the target outcome (reward) or at the state input (eg., cues) as outcome-specific and feature-specific prediction errors. Distributional temporal difference learning (TD) models have been used to model both sources of heterogeneity in DA responses [14, 15]. This distributional approach can be extended to spatial representations, such as the activity of hippocampal place cells, offering a potential explanation for variations in place cell behavior that have remained elusive in SR models.

In summary, DRL provides a more biologically plausible framework by modeling variability in rewards, allowing for the parallel processing of uncertainty in complex environments.

## 2.3 The Distributional Successor Measure

The Distributional Successor Measure (DSM) combines the principles of distributional reinforcement learning and successor representations to capture the diversity of future outcomes an agent may experience. Rather than representing the expected number of times a state will be visited in the future, the DSM models the full distribution over these future visitations. This richer representation allows the DSM to capture not only the expected future occupancy but also the variability and uncertainty inherent in the environment.

Formally, the DSM, represented as  $\bar{\Psi}^\pi(x)$ , maps each state  $x$  to a distribution over the “random discounted state-occupancy measure”  $M^\pi(S|x)$ , which describes the probability that, starting from state  $x$ , the agent will visit a measurable set of states  $S \subseteq \mathcal{X}$  over time, discounted by a factor  $\gamma \in [0, 1)$ .

The DSM improves over traditional models by learning the full distribution of these occupancy measures, allowing it to account for both the aleatoric uncertainty in the environment and the stochastic dynamics of the policy. This mapping is updated recursively through the distributional Bellman equation:

$$\bar{\Psi}^\pi(x) := \mathbb{E}_{X' \sim P^\pi(\cdot|x)} \left[ (1 - \gamma)\delta_x + \gamma\bar{\Psi}^\pi(X') \right],$$

where  $\delta_x$  is the Dirac delta function centered at the current state  $x$ , and  $\gamma$  is the discount factor.

Due to the high dimensionality of the state space, representing the full distribution  $\bar{\Psi}^\pi(x)$  is computationally infeasible. To address this, DSM uses an Equally-Weighted Particle (EWP) representation to approximate the full distribution over future states. In this representation, the distribution is approximated by a finite set of Dirac masses or delta functions, each centered on a specific point in the state space. This is formalized as:

$$\bar{\Psi}^\pi(x) \approx \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x)},$$

where each Dirac mass  $\delta_{\theta_i(x)}$  represents a particle or atom in the state space, and  $\theta_i(x)$  are the parameters of the distributional particles that encode the future state distributions.

This allows the model to approximate the full distribution using a tractable number of components. The particles  $\theta_i(x)$  represents a probability distribution over a potentially large state space and are implemented as generative models, much like the aforementioned  $\gamma$ -models, using deep neural networks that take in noise variables and output the corresponding future state distributions.

Training the DSM model involves minimizing a loss based on Maximum Mean Discrepancy (MMD), which compares the learned distribution  $\bar{\Psi}^\pi(x)$  to its target distribution using a kernel-based distance measure that computes the difference between the distributions in a high-dimensional space. The generator in the model works to min-

imize the MMD loss, while the discriminator maximizes it, following an adversarial training framework.

In addition to MMD loss, n-step bootstrapping is used during training to improve stability and accelerate convergence. This technique allows the model to consider multiple future steps of future transitions when updating the DSM, improving its ability to learn long-term dependencies in the environment. An adaptive kernel is employed to adjust the MMD loss as the model evolves, ensuring that the kernel remains effective as the state distributions change.

A unique feature of the DSM is its ability to estimate the full distribution of returns, capturing the aleatoric uncertainty—uncertainty arising from the inherent randomness of the environment and the stochastic nature of the MDP dynamics. While traditional models focus on expected future rewards, DSM provides a complete view of the potential future states and outcomes, offering more statistical information than scalar predictions. This makes DSM particularly useful in decision-making processes where knowledge of all the possible outcomes allows for more informed planning, especially in scenarios where minimizing risk is crucial.

### **Rationale behind using distributed SR to model place cells:**

In distributing future state representations across multiple predictive components (atoms), the DSM may better model how biological systems, such as the hippocampus, handle navigation tasks. Traditional SR models are limited in their ability to adapt quickly to changes in transition structures or policies, but the DSM’s distributed nature allows for different atoms to encode different transition dynamics, thus enabling more context-sensitive decision-making.

This flexibility may mirror the way the hippocampus and associated regions update internal representations based on new experiences. Mechanisms such as experience replay—where sequences of states are reactivated during sleep or rest—support the idea that hippocampal navigation is inherently distributional, further grounding the DSM in biological processes.

In summary, by leveraging DSM’s distributed representation, models of place cells should be able to account for the complex, interconnected structure of the environment,

enabling more accurate predictions of future trajectories. Place cells, which represent different parts of an environment, show variability in their activity that DSM should capture more effectively than SR. This allows DSM to more accurately predict future trajectories in complex environments, further supporting its role as a powerful model for both artificial and biological navigation systems.

# Chapter 3

## Methods and Experiments

In this chapter, we describe the two main environments utilized in this study; the inverted pendulum task and a simulated rat-foraging environment, and the experimental setup for each environment (dataset generation). We detail the architecture of the Distributional Successor Measure (DSM) model and the parameters used for its training. We then describe the neuroscientific tools employed to interpret the model’s internal representations and investigate how DSM may generalize to novel environments.

### 3.1 Environments and Experimental Setup

We conduct our experiments in two environments that differ in state space dimensionality, stochasticity, and dynamics.

#### 3.1.1 Inverted Pendulum Task

The inverted pendulum task [63, 64] from OpenAI Gym serves as a basic, well-studied reinforcement learning environment to analyze how the model captures control tasks with no apriori knowledge of the dynamics. The system, as shown in Figure 3.1, comprises a pendulum with one end fixed in place and the other end free to move.

The pendulum starts in a random position, and the goal is to swing it into an upright position with its center of mass directly above the fixed point. The task requires applying forces to keep the pendulum upright, with torque applied counterclockwise. The default reward function penalizes deviations from an upright position and excessive

velocities, with the maximum reward being 0 and the minimum being -16. The state  $s_i$  is given by the pendulum's orientation  $\theta \in (0, 2\pi)$  and its angular velocity  $\dot{\theta}$  in the range  $(-8, 8)$  with the observations as 3 dimensional vectors consisting of the cosine and sine of the angle and its angular velocity.

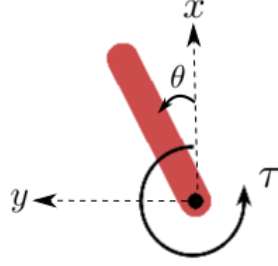


Figure 3.1: Diagram of the inverted pendulum system, showing the coordinate system with orientation  $\theta$ , applied torque  $\tau$ , and the horizontal and vertical axes. The pendulum rotates around the fixed pivot point

Dataset generation: A Soft Actor-Critic (SAC) policy [65] is trained on the environment with the default reward for 500,000 steps, optimizing the agent's actions to control the pendulum's movement efficiently. After training, the learned policy is used to interact with the environment, generating a dataset that consists of 500,000 transitions consisting of step type and observations. The dataset is collected by evaluating the trained policy over multiple episodes.

In this setup, the DSM model learns to predict the distribution of future pendulum angles and velocities based on each source state from the dataset.

### 3.1.2 Simulated Rat-Foraging Environment

To model hippocampal activity and spatial navigation, we simulated an agent moving within a 1m x 1m 2D navigation environment using the RatInABox simulation package [66]. This toolkit provides a biologically grounded framework for simulating rodent foraging behavior and generating associated neural data from spatially selective cells, such as place cells, within continuous, complex surroundings.

We configured the environment with a random cover of 50 place cells, each with a Gaussian place field of width 0.2m, randomly distributed across the environment, as shown in Figure 3.3. Each place cell has a preferred firing location, and its firing rate

decreases with distance from this center according to a Gaussian distribution. This is illustrated in Figure 3.2, where the rat’s trajectory, place cell rate maps, and corresponding activity timeseries are depicted, demonstrating encoding of spatial locations and variation of place cell activity with agent’s position.

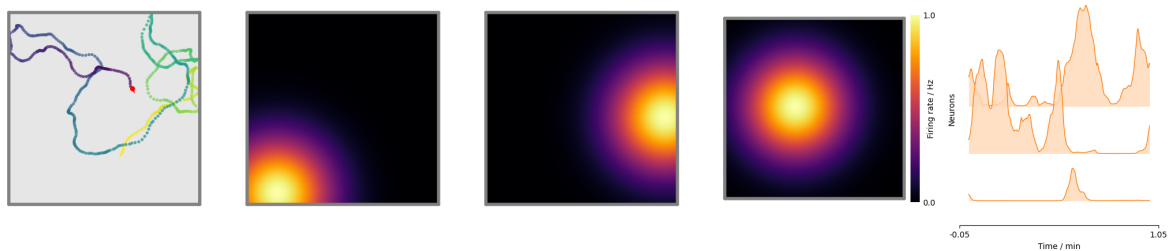


Figure 3.2: Illustration of a rat’s trajectory (left), three place cell rate maps (middle), and place cell activity timeseries (right)

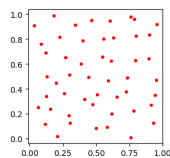


Figure 3.3: Place cell centres randomly distributed in an environment

To capture a range of navigational behaviors, we employed four distinct movement policies, simulating 20 minutes of activity for each:

- **Random Walk:** The agent performs a random walk governed by Ornstein-Uhlenbeck processes [67] across the environment, with default parameters based on rat locomotion data [68]. In this policy, the agent moves stochastically, simulating exploratory foraging behavior without a specific goal.
- **Teleportation:** The agent randomly teleports between locations within the environment, disrupting spatial continuity and providing a scenario where typical path integration cues are absent.
- **High Thigmotaxis:** The agent exhibits wall-hugging behavior, a pattern often associated with anxiety-like states in rodents. This movement policy biases the agent to stay close to the boundaries of the environment.

- **Goal-Directed Movement:** The agent navigates toward a central goal location, guided by a reward signal, simulating purposeful movement commonly observed when rodents are trained to seek rewards.

The neural activity generated from these different movement policies forms the basis for understanding how spatial information is encoded across varying behavioral contexts. Instead of recording the agent’s Cartesian coordinates, our datasets comprise the activities of all place cells as the agent traverses the environment.

To evaluate the DSM model’s ability to handle spatial constraints, we also generated a dataset where the agent moves randomly in a 1m x 1m environment containing a wall placed near the middle. The wall blocks most of the path, leaving only a 0.05m gap at the top for the agent to navigate through. This setup introduces spatial constraints that limit the agent’s movements and affect the firing of place cells representing the agent’s location. The main purpose of this dataset was to test whether the DSM can adeptly learn from place cell activities to predict future states that respect these boundaries.

The trajectories of the datasets can be seen in Figure 3.4

### **Position decoder**

To decode the agent’s position from the activity of place cells, we employed a dropout-based neural network. This network was designed to map neural firing rates from place cells to its environmental coordinates, allowing for the reconstruction of the agent’s position based on its internal representation of space.

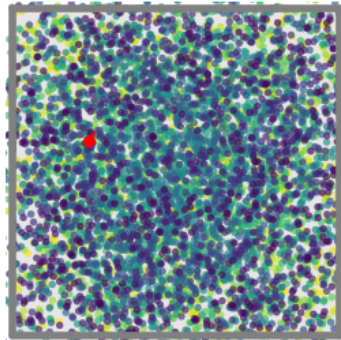
The network architecture consisted of:

- **Three fully connected layers:** An input layer of size 50 corresponding to the number of place cells, two hidden layers with 128 and 64 units respectively, and an output layer of size 2 predicting the agent’s x and y coordinates.
- **Dropout layers:** Applied after each hidden layer with a dropout probability of 0.3 to introduce stochasticity during training, allowing the network to capture uncertainty in the position estimates.

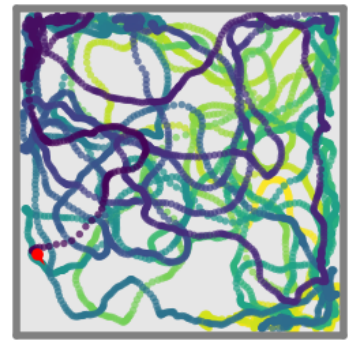
The environment was discretized at a high resolution (0.01m), creating a grid of possible positions for the agent. The position decoder, also referred to as dropoutNet



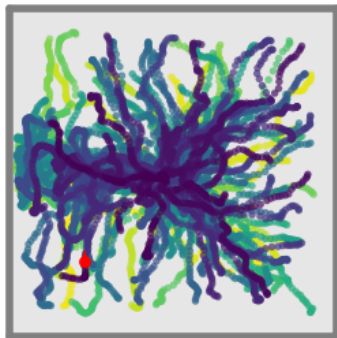
**(a)** Random walk policy trajectory



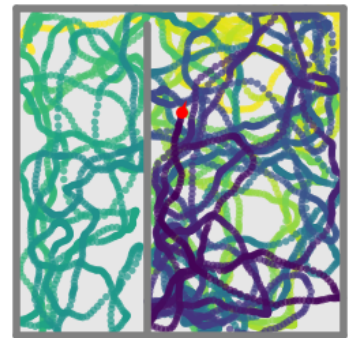
**(b)** Teleport policy trajectory



**(c)** High Thigmotaxis behavior trajectory



**(d)** Goal-directed trajectory



**(e)** Trajectory constrained by walls

Figure 3.4: Simulated RatInABox trajectories: (a) random walk (b) teleportation (c) high thigmotaxis behavior (d) goal-directed movement and (e) wall constrained agent's movements

was trained using simulated firing rates extracted for each grid point, enabling it to learn the mapping from place cell activities to spatial coordinates.

In this setup, the DSM model is tasked with predicting the distribution of place cell activities that encode predictive maps, conditioned on the agent’s movement policy given in each dataset. By modeling the full distribution of future place cell activities, the DSM captures the uncertainty and variability in the agent’s future locations, providing a more nuanced understanding of how the hippocampus may encode spatial uncertainty.

### 3.2 DSM Model Network and Training Parameters

We implemented the Distributional Successor Measure (DSM) model with 10 generative models  $\theta_i(x)$ , referred to as *model atoms* or *state distributions*. Each atom represents a probability distribution over the state space  $\mathcal{X}$ . These atoms are modeled by multi-layer perceptrons (MLPs), where each MLP consists of three hidden layers with 32 neurons in each layer, using Leaky ReLU as the activation function throughout.

Each generative model receives an input that is a concatenation of the current state and a random latent vector. The state dimension varies depending on the environment – for example, 3-dimensional states in the Pendulum environment and 50-dimensional states in the Ratinabox simulations. Additionally, each atom receives a random latent vector of 5 dimensions, enabling it to model variation in future state predictions. The outputs of each atom represents its predicted successor representation, mapping each source to a distribution over future states.

We trained a total of six DSM models from all the datasets generated with the pendulum and RatInABox environments. All models were trained for 500,000 gradient steps, with a batchsize of 32 samples.

| Hyperparameter               | Configuration                                                  |
|------------------------------|----------------------------------------------------------------|
| Number of atoms              | 10                                                             |
| State dimensions             | Environment-dependent (e.g., 3 for Pendulum, 50 for RatInABox) |
| Latent dimensions            | 5                                                              |
| Hidden layers                | 3 layers, each with 32 neurons                                 |
| Activation function          | Leaky ReLU                                                     |
| Discount factor ( $\gamma$ ) | 0.95                                                           |
| Bootstrapping horizon        | 6 steps                                                        |
| Training steps               | 500,000                                                        |

Table 3.1: Hyperparameters and configuration for the DSM model.

For training, we employed the configurations outlined in Table 3.1. All other parameters are set to the default values outlined in the base DSM paper [12]. We used Weights & Biases (WandB) for logging metrics and Orbx checkpoints for saving model states, ensuring reproducibility and monitoring of progress during the training runs.

### 3.3 Interpretability Techniques

To gain insight into the internal workings of the DSM model, we applied several neuroscientific analysis techniques to the model’s learned representations:

#### 3.3.1 Pendulum experiments

We visualized the activations of each atom’s neurons in the trained DSM model across two dimensions: orientation ( $\theta$ ) and angular velocity ( $\dot{\theta}$ ), using contour plots to show how neurons responded to different parts of the state space. These 2D maps offer insight into the neural representations formed by atoms of the DSM model, revealing how neurons encode different state variables, thus aiding in interpretability.

#### Entropy Analysis

Neuroscientists often use entropy methods to quantify neuron selectivity by measuring the uncertainty or variability in neuronal activation patterns, revealing how distinct or overlapping these patterns are in response to different stimuli or brain states [69]. Wang,

Longwei, et al. has shown that the randomness in neuron activation patterns decreases with depth, indicating that deeper layers exhibit more stable activations compared to shallower ones, and we seek to investigate this in the model. [70]

We computed entropy across the neurons in each layer of the DSM with respect to specific input dimensions of  $\theta$  and  $\dot{\theta}$  to quantify how information is distributed within the network. This would help identify neurons that encode high-level, invariant features of the environment, as well as those that are sensitive to subtle changes in the source states. High entropy suggests that a neuron’s activations fluctuate significantly across different inputs, indicating sensitivity to a wide range of stimuli and resulting in diverse, less predictable outputs. Conversely, low entropy reflects more uniform activations in response to various inputs. In the extreme case, if the neuron’s output remains nearly constant across all velocities, it indicates a highly predictable response, leading to low entropy.

### **Dimensionality Reduction:**

We employed two dimensionality reduction techniques: t-distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) to project the high-dimensional internal activations of the DSM into lower-dimensional spaces.

t-SNE [71] is a nonlinear method that minimizes the divergence between two distributions: one that measures pairwise similarities of the input high-dimensional data points and one that measures pairwise similarities of the corresponding low-dimensional points in the embedding. It excels at preserving local structure and revealing clusters formed by similar activations.

UMAP [72] is based on manifold learning and algebraic topology. It assumes that the data is uniformly distributed on a Riemannian manifold and that this manifold is locally connected. UMAP preserves both local and some global structure of the data, and it is computationally efficient for large datasets.

By applying t-SNE and UMAP to the activations of the DSM’s intermediate layers, we could observe how different variables—representing state space dimensions or environmentally relevant features—are mapped within the network. This analysis may help us understand the model’s internal encoding mechanisms.

## Linear Decoder Analysis

One component of our interpretability analysis was the use of a linear decoder to map the activations of internal neurons to relevant state variables. Two Ridge regression models were trained: one to predict source states and another to predict future states from the internal activations of a DSM atom. Their performance was evaluated under various conditions to assess the atom’s ability to encode task-relevant information.

First, we split the environment sources into training and test sources. We selected a random latent vector, generated samples from the atom for each source state, and collected activations from the intermediate layers. The generated sample, representing the state of the environment after the agent takes an action, is referred to as the “end predicted state”. The data was split into two sets: ‘source states - internal activations’ and ‘end predicted states - internal activations,’ which were used to evaluate the atom’s ability to represent both current and future states.

To further probe the model’s behavior, we introduced a new latent vector and re-computed the activations across the same layers for both the training and test data. This allowed us to observe how the model responds when tested with a latent configuration different from the one used during training, helping us identify how well the internal activations of the DSM model reflect the actual and predicted states of the environment irrespective of the random input latent vectors.

**Decoding Source States:** We examined how the model’s internal neurons encode current state information, such as the pendulum’s orientation or the rat agent’s spatial location. A Ridge regression model was trained to map internal activations to source states in the training set using the original latent vector. Performance was evaluated using the  $R^2$  score under three conditions: (1) the same latent vector but different source data (test set), (2) a different latent vector with the same training source, and (3) both a different latent vector and different source data.

**Decoding Predicted States:** We repeated the procedure, training a separate Ridge regression model to map intermediate activations to end predicted states using the original latent vector.

These analyses provide insights into the DSM atom’s capacity to encode and predict environment-relevant variables, offering a clearer understanding of how the model

generalizes across different latent configurations and state variables.

### **3.3.2 Place cell experiments**

We repeated the entropy analysis, dimensionality reduction, and linear decoder analysis for the RatInABox environment to identify correlations between model activations and source states.

#### **Comparison of learnt representations across movement policies**

Different movement policies can generate diverse representations of the environment, with random walks producing fundamentally different predictive maps compared to more structured, goal-directed behaviors. For instance, random walks might lead to more symmetrical rate maps, reflecting an even exploration of the environment, whereas policies like thigmotaxis, which involve boundary-hugging behaviors, could result in asymmetrical maps with higher concentrations near the edges.

To quantify these differences, we will compare the spatial structure of the learned rate maps using metrics like entropy, which can capture the spread and variability in the representations. Additionally, we will employ t-SNE to visualize how distinct the internal representations are across different movement policies, projecting the learned state embeddings into two-dimensional space [73]. This approach will help us better understand how the DSM model organizes its predictive maps based on varying behavioral strategies and whether its internal representations reflect key features of the environment and movement patterns.

#### **Evaluating DSM’s Generalization Capabilities**

A key objective of this study was to evaluate the robustness of the DSM model in adapting to environmental changes. We simulated environmental alterations by incrementally changing the positions of 1 to 50 place cell centers (PCs), randomly reassigning them within the environment. We then input the firing activities of these altered PCs into the DSM model to evaluate how well it could adapt to these perturbations.

In real-world experiments, this approach is analogous to modifying environmental cues—such as changing the color of surroundings or introducing obstacles—which typ-

ically affects only a subset of cues [35]. We hypothesized that the DSM would align with biological observations, where most place cell rate maps remain stable under minor environmental shifts but exhibit global remapping when larger-scale changes occur.

Assuming the agent follows a uniform policy without directional bias (represented by different atoms), we averaged the rate maps across all 10 atoms for each place cell, resulting in 50 rate maps. These maps served as the DSM’s predicted representations, which we then compared to the modified maps after place cell centers were altered. Visual representations of these comparisons are presented in the Results section.

To benchmark the DSM’s performance, we trained the traditional Successor Representation (SR) model on the same dataset with identical place cell center configurations. For both the DSM and SR models, we quantified the degree of change in place cell representations before and after the environmental modifications by calculating similarity measures (cosine similarity) between the original and modified predicted rate maps of the 50 place cells. To ensure statistical robustness, we applied bootstrapping by reassigning random place cell centers up to three times for each level of change. To ensure a fair comparison, a fixed random seed was used, guaranteeing consistent place cell center positions across both models.

To conclude, this experimental framework allowed us to systematically evaluate the DSM model’s capacity to maintain stable representations amidst gradual environmental changes, providing key insights into its generalization capabilities and adaptability compared to the traditional SR model.

# Chapter 4

## Results and Discussion

In this chapter, we illustrate some of the learnt representations of the model in both environments, and show the results of using neuroscientific tools to interpret the model’s internal representations. We also show how the representations may change with different policies as investigated by the different datasets in the RatInABox environment, as well as the results of our experiments to compare the SR and the DSM models. We start with showing our results on the Pendulum environment, and then the RatInABox experiments, before the comparison of SR and DSM.

### 4.1 Pendulum experiments

The successor samples generated by the DSM model after training on the Pendulum dataset are shown in Fig. 4.1. Individual atoms of the model generate separate samples based on the source state (shown by the red “x” mark in the figure).

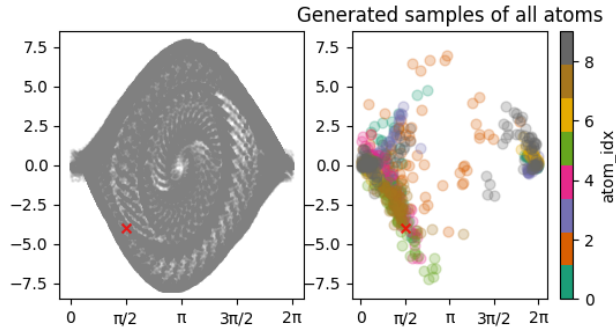
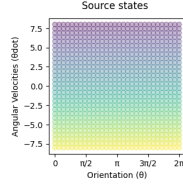
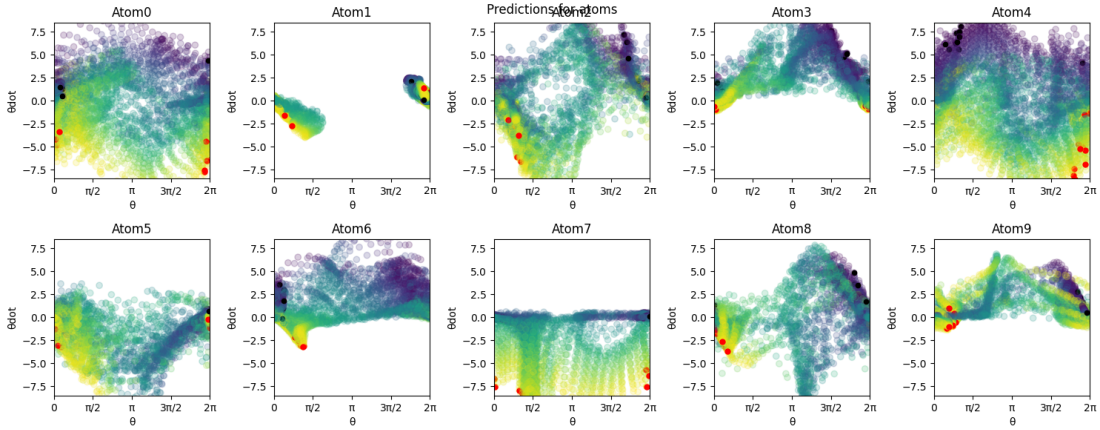


Figure 4.1: Left: Pendulum dataset generated by the SAC policy. Right: DSM generated samples (32 per atom) for source ( $\theta = \frac{\pi}{2}, \dot{\theta} = -4$ ).



(a) The grid of source states representing the pendulum's angle  $\theta$  and angular velocity  $\dot{\theta}$  in the state space



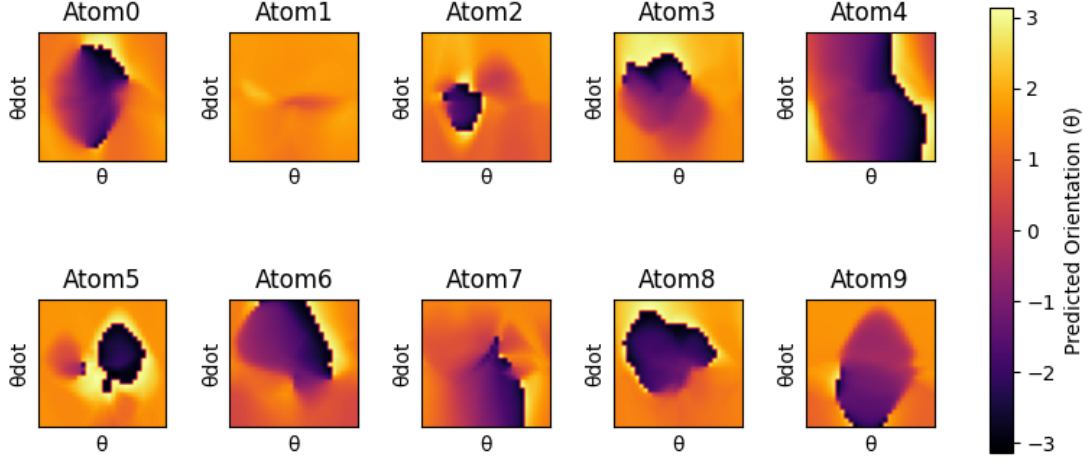
(b) DSM predictions from each atom for pendulum states

Figure 4.2: DSM predictions from each atom

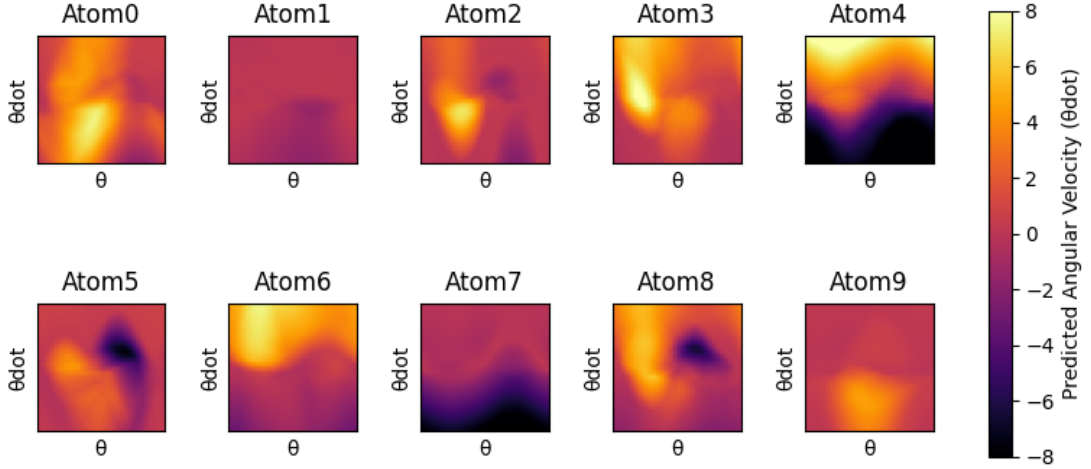
We divided the input space into 900 discrete states and generated ten samples from the DSM atoms for each source state, as shown in Fig. 4.2. This was done to assess whether the atoms exhibit any bias toward specific regions of the environment (source states or observations). These atoms collectively offer a more nuanced prediction than a single point estimate. The visualizations show that each atom captures distinct aspects of the state space, with some atoms predicting broader, more variable outcomes and others concentrating on narrower, more specific behaviors. For example, Atoms 6 and 7 appear to encode for very specific range of angular velocities, indicating a sensitivity to particular dynamic behaviors. In contrast, Atom 1 seems to show very localized behavior, primarily predicting the stable equilibrium points of the pendulum.

Next, we computed the intermediate neuron activations in the DSM model. We first present the rate map of the final layer for each atom in relation to the discretized source states (Fig. 4.3). Figure 4.3b shows the angular velocity predictions, while Figure 4.3a

converts the activations from neurons predicting the sine and cosine of the angle into the corresponding values of orientation  $\theta$ .



(a) Orientation  $\theta$  predictions



(b) Angular velocity  $\dot{\theta}$  predictions

Figure 4.3: DSM predictions of each atom over  $\theta$  and  $\dot{\theta}$

To examine and illustrate, we choose atom 7 for further visualizations. From the above figure, we can see that it shows angular velocity predictions mostly in the range  $(-8,0)$ . We illustrate how the neurons in the middle layer (Dense 1) of atom 7 respond to different regions of the state space in Figure 4.4.

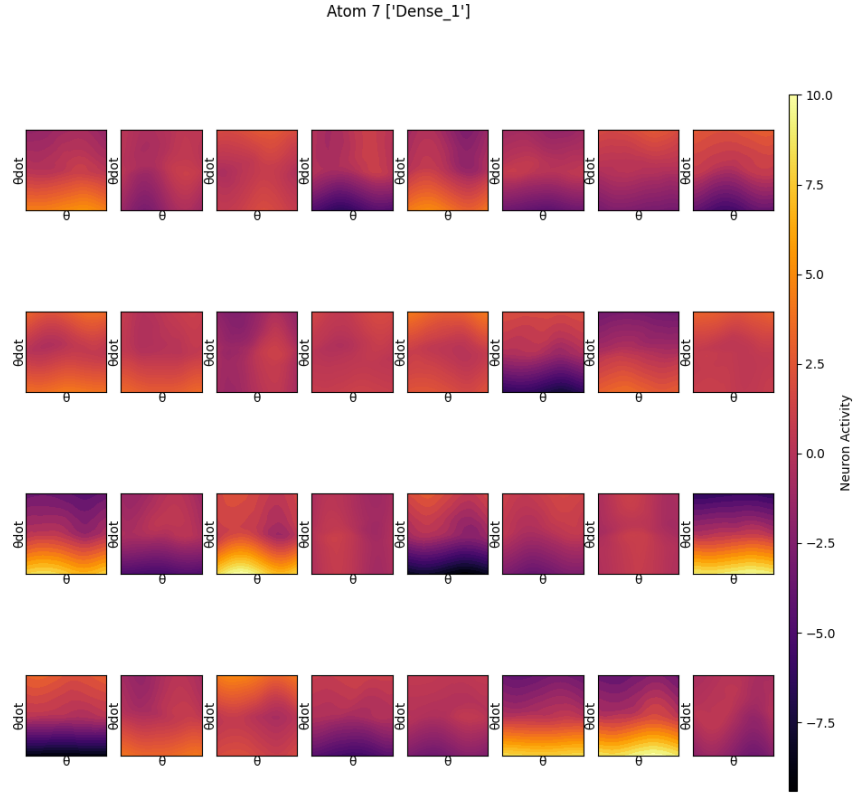


Figure 4.4: Atom 7: Activations of the Dense1 layer over source states

We also examined the influence of setting different latent seeds on the generated representations of the model (Atom 7) in Figure 4.5. This should provide an insight into the variety of samples generated by the atom.

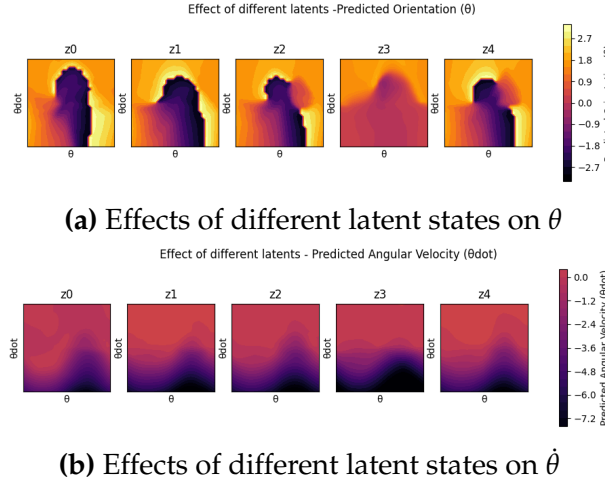


Figure 4.5: Effects of different latent states

## Entropy Analysis

In our entropy analysis, we observed the distribution of entropy values for neurons across different layers of the network, split by their contribution to encoding angular and velocity information, as shown in Fig. 4.6.

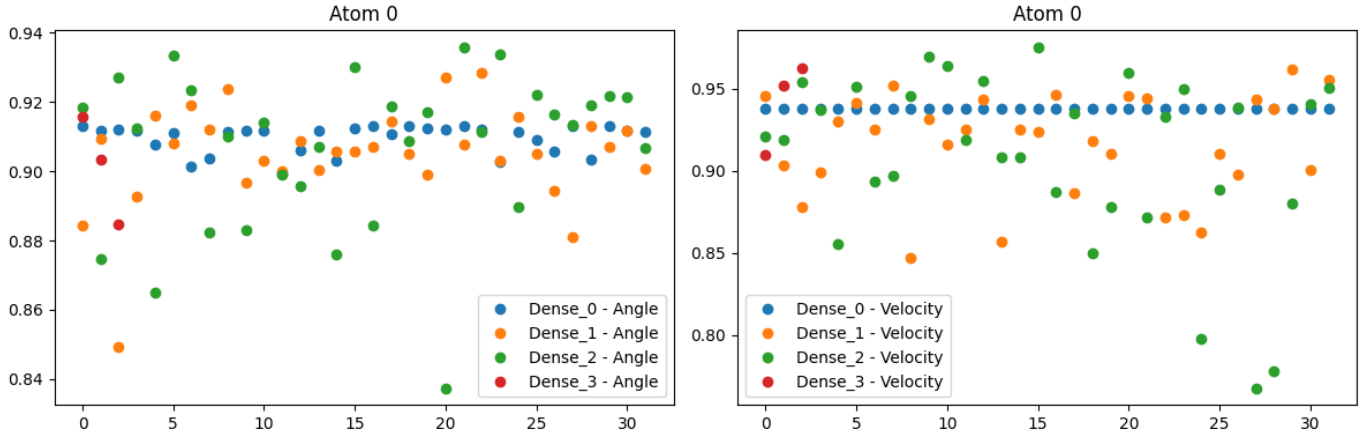


Figure 4.6: Entropy values for Atom 0

The entropy values appear to be distributed across layers in a relatively similar manner, suggesting that neurons in both early and later layers contribute comparably to encoding these features. Interestingly, the neurons in the first layer of each atom exhibit

similar levels of entropy when encoding velocity, indicating that early layers process velocity information uniformly. Rather than showing increased flexibility in higher layers, the results point to a distributed encoding across the network, with neurons in all layers involved in processing both angular and velocity information. This distributed activation pattern implies that the network integrates inputs nonlinearly across layers, without a clear hierarchical abstraction as one might expect. The lack of significant variation in entropy across layers suggests that there is no strict disentanglement of angle and velocity information, similar to how the brain often encodes multiple features simultaneously within the same neural populations.

This finding mirrors the neural behavior observed in some neurons, where neural responses are not strictly separated by input type. For example, while place cells in the hippocampus encode spatial locations some of them also represent other dimensions such as running speed and head direction [74,75]. Across various brain regions, neurons exhibit mixed selectivity, responding to a combination of factors rather than isolated inputs [76].

### **Linear Decoder Analysis**

Our primary goal was to understand how well the neurons within the DSM model encode relevant environmental information and how this encoding generalizes across different latent representations and data sources. We evaluated the DSM model’s ability to decode current source states and predicted future states from its internal activations using  $R^2$  scores across different layers. The analysis was conducted under three conditions: (1) same latent vector, different source; (2) different latent vector, same source; and (3) different latent vector and source, to assess the model’s encoding and generalization capabilities. By calculating the  $R^2$  scores for each condition across layers and individual neurons, we aim to understand how the DSM model’s internal representations evolve across layers and how well these representations capture both current and future state information.

#### **Decoding Source States:**

Table 4.1: Layers: R2 Scores for different Latent and Source Combinations

| Layer   | Original latent - different source | Different latent - same source | Different latent - different source |
|---------|------------------------------------|--------------------------------|-------------------------------------|
| Dense 0 | 0.999                              | 0.878                          | 0.876                               |
| Dense 1 | 0.998                              | 0.751                          | 0.746                               |
| Dense 2 | 0.995                              | 0.723                          | 0.720                               |
| Dense 3 | 0.680                              | 0.548                          | 0.526                               |

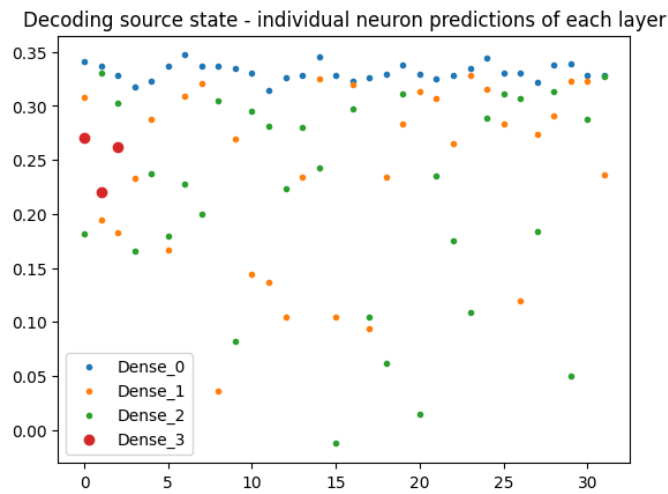


Figure 4.7: R2 scores in decoding test source states (with same latent) from individual neurons in different layers

### Decoding Predicted States:

Table 4.2: Layers: R2 Scores for Different Latent and Source Combinations

| Layer   | Original latent - different source | Different latent - same source | Different latent - different source |
|---------|------------------------------------|--------------------------------|-------------------------------------|
| Dense 0 | 0.761                              | 0.355                          | 0.277                               |
| Dense 1 | 0.989                              | -0.829                         | -0.982                              |
| Dense 2 | 0.992                              | 0.882                          | 0.869                               |

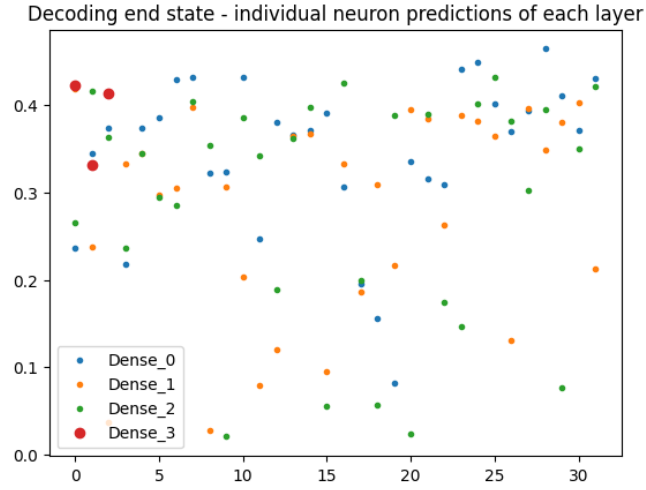


Figure 4.8: Individual neurons of different layers: R2 scores in decoding end states (with same latent)

The  $R^2$  scores showed that the model predicts current and future states well when operating under varying latent space and input source conditions. For decoding source states, earlier layers (e.g., Dense 0 with  $R^2 = 0.999$ ) captured most relevant information, when tested with a different latent vector but the same source data, the results reflect a more noticeable drop in accuracy. For instance, Dense 0 still maintains a relatively high score of 0.878, but the deeper layers such as Dense 3 experience a more significant decline to 0.548. This suggests that deeper layers are more sensitive to changes in the latent representation, possibly because the information becomes more entangled or abstract as it passes through the network. However, the DSM model shows strong predictive performance for future states with deeper layers, showing its robustness in predicting future outcomes based on internal activations. Since the 'Dense 3' layer corresponds to the output, we omit it from Table 4.2

### Dimensionality Reduction:

We employed t-SNE and UMAP to visualize how different dimensions of our state space may be encoded in the model's hidden layers. The representations are shown in the figures below.

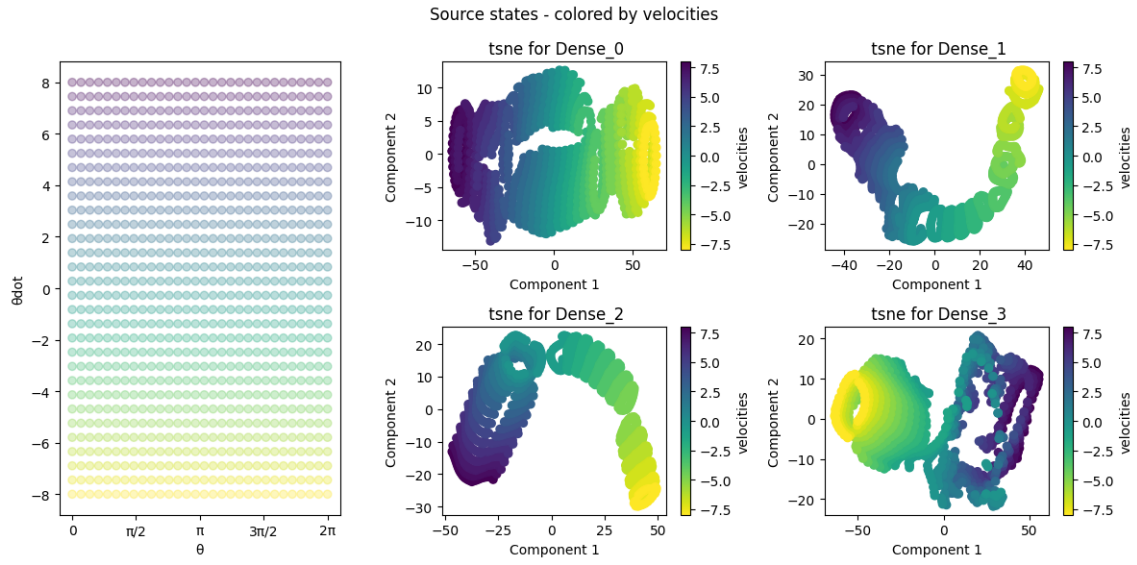


Figure 4.9: 2 component t-SNE coloured by increasing order of velocity

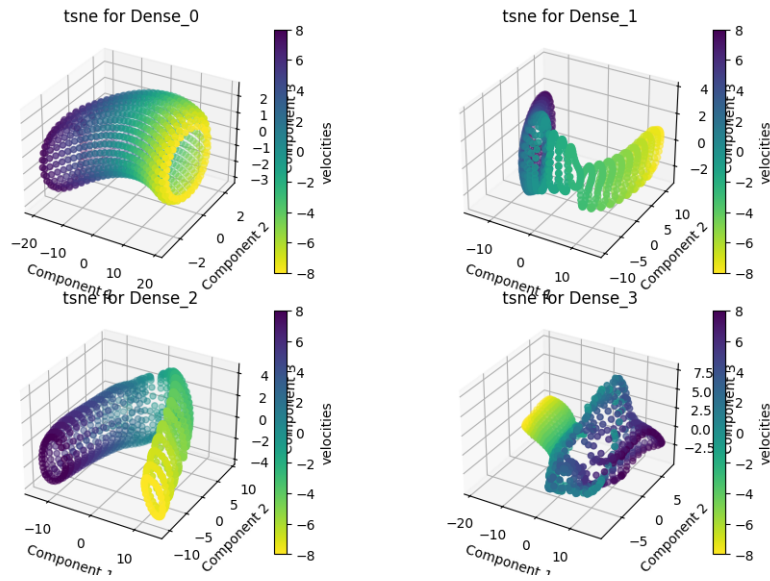


Figure 4.10: 3 component t-SNE coloured by increasing order of velocity

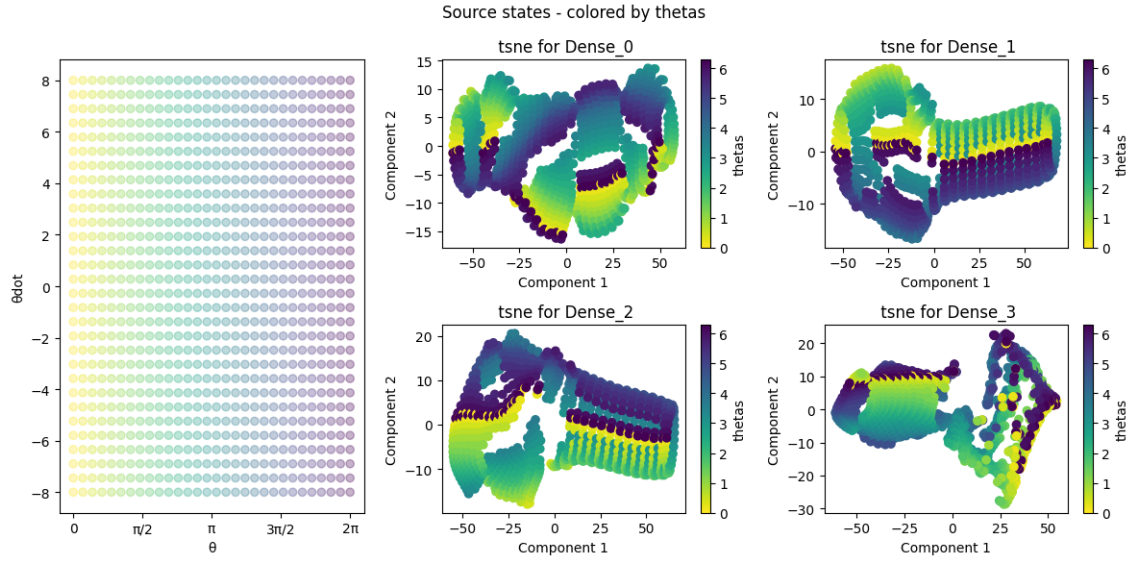


Figure 4.11: 2 component t-SNE coloured by increasing order of theta

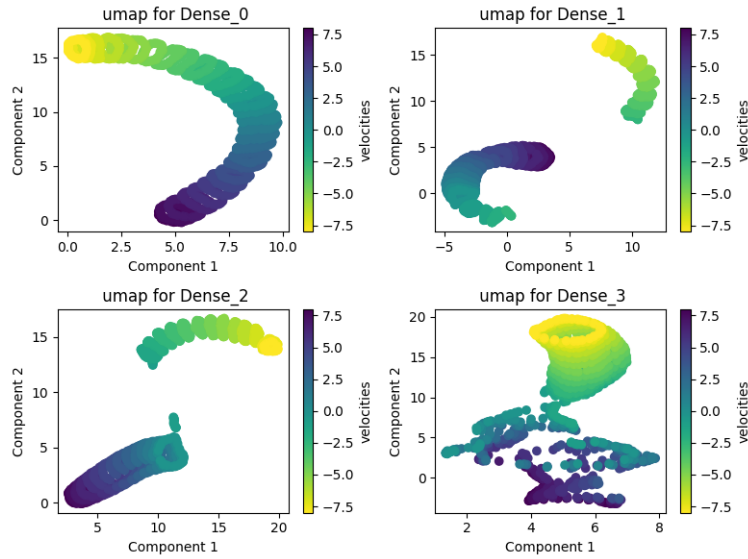


Figure 4.12: 2 component UMAP coloured by increasing order of velocity

We observe that the DSM organizes states in each layer in a very order manner according to the environment's dynamics, with similar environmental features arranged together and dissimilar features positioned farther apart. The UMAP also shows clustering for low and high velocities for the second and third layers.

## 4.2 Place cell experiments

In this section, we present the results of our experiments involving place cells in the simulated rat-foraging environment. We focus on how the DSM model represents place cell activities in different movement policies and how it compares with the SR with respect to environmental changes caused by modifying place cell centres.

We first demonstrate that the DropoutNet can accurately reconstruct the agent's positions from the input place cell activity. Figures 4.13a and 4.13b show the decoded positions from the wall dataset and the random walk dataset, respectively.

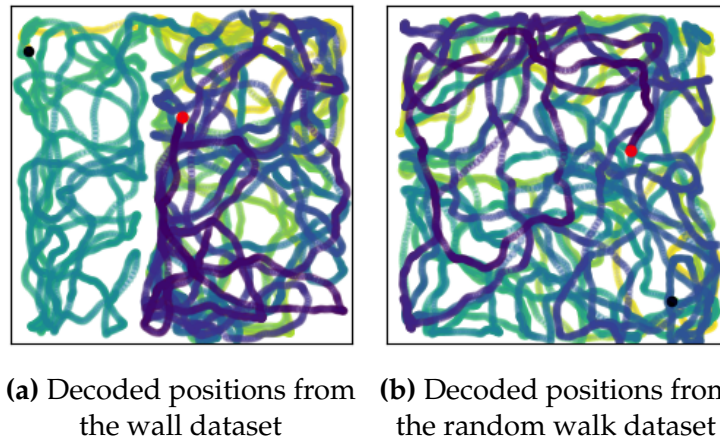


Figure 4.13: Comparison of decoded positions from two datasets using DropoutNet. (a) Wall dataset. (b) Random walk dataset.

Next, we analyze the place cell activations that serve as inputs to the DSM model. We take the random walk dataset and compute the rate maps, which show the predictive representations of the input place cells (PCs). Figure 4.14 visualizes the rate maps of five randomly selected place cells.

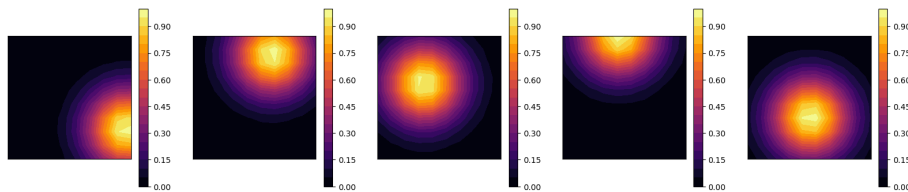


Figure 4.14: Input place cell rate maps for five randomly selected PCs from the random walk dataset.

To show that the place cell activations are evenly distributed across the environment, we compute the average activity over all place cells. This is illustrated in Figure 4.15.

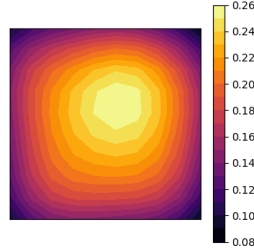


Figure 4.15: Average place cell activity across all PCs in the random walk dataset, showing even distribution.

While we have 10 atoms that encode different trajectories, we assume a uniform policy by averaging the activity across the different atoms to extract a predictive successor representation (SR) of the place cells. Figure 4.16 shows the averaged predictions of the rate maps for the five place cells selected above. These rate maps are from the DSM model trained on the random walk policy.

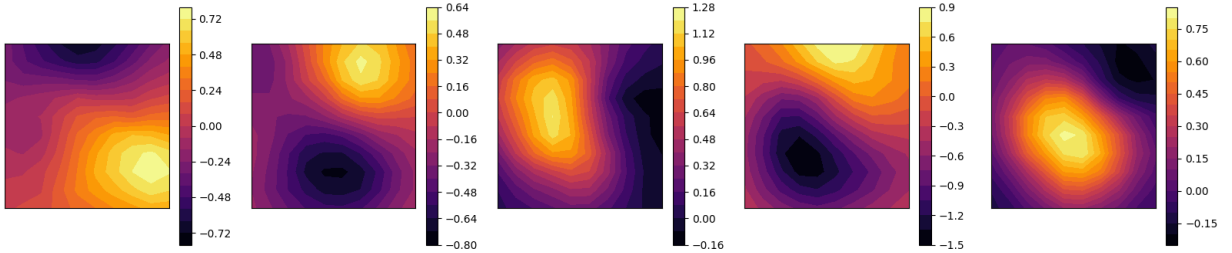


Figure 4.16: Averaged predicted rate maps for the same five selected place cells, obtained by averaging across all atoms under a uniform policy assumption.

Figure 4.17 illustrates the predicted rate map of a single place cell across all atoms, highlighting how each atom contributes to the overall prediction. These rate maps are from the model trained on the random walk policy.

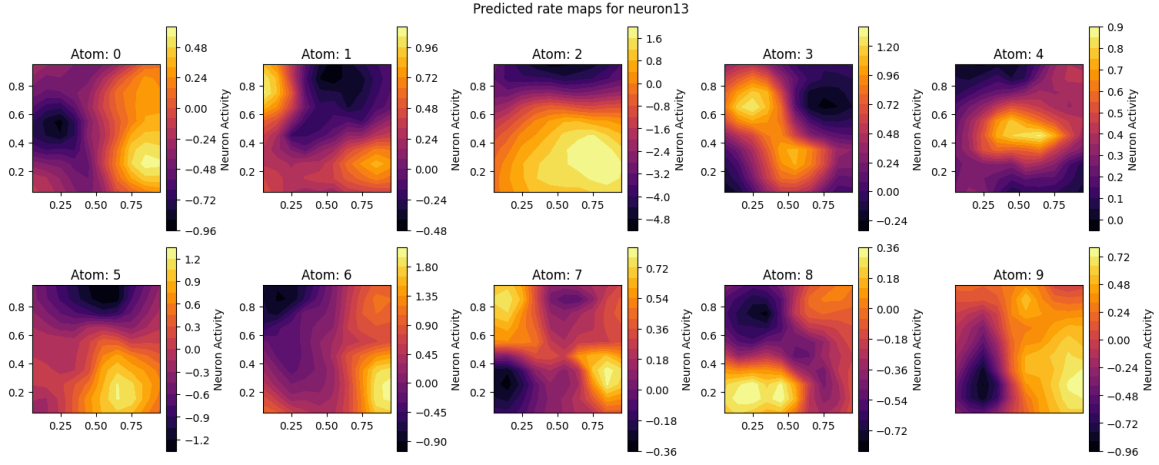


Figure 4.17: Predicted rate maps of an individual place cell across all atoms.

### Comparison of Learned Representations for Different Movement Policies

We compared the DSM's learned representations under different movement policies to understand how policy influences the predictive maps.

Figure 4.18 shows the DSM's predicted successor samples for the random walk policy. It displays a single sample produced by each atom for multiple source states, while Figure 4.19a focuses on multiple source states produced by each atom for a single source state.

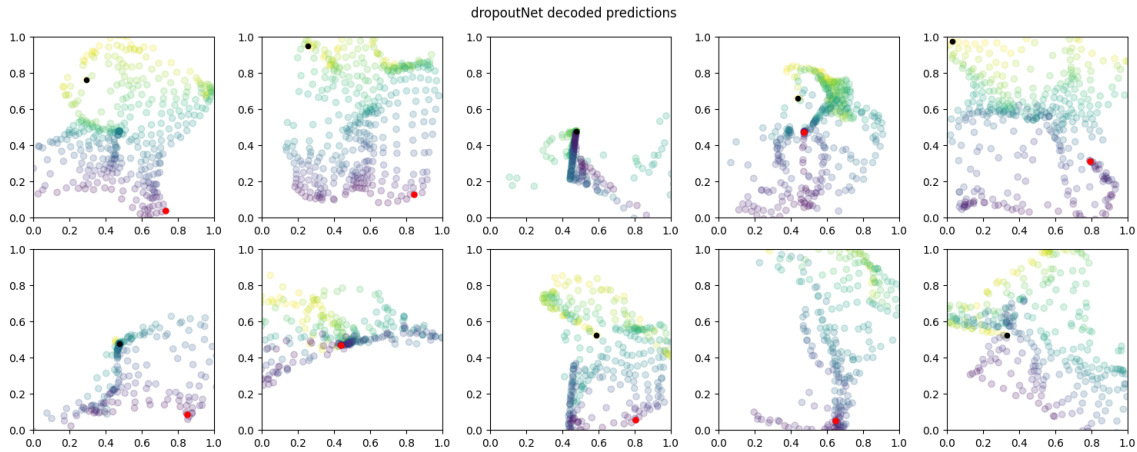


Figure 4.18: DSM predicted successor samples under the random walk policy for multiple source states - each subfigure represents a different atom

Figure 4.19 illustrates the predictions for a single source state from DSM models trained on different policies. While the predicted samples tend to be highly biased

towards the policy they were trained on, individual atoms (represented by different colors) make distinct predictions. However, in the teleport policy model, the atoms' predictions appear more scattered and do not follow any clear clustering.

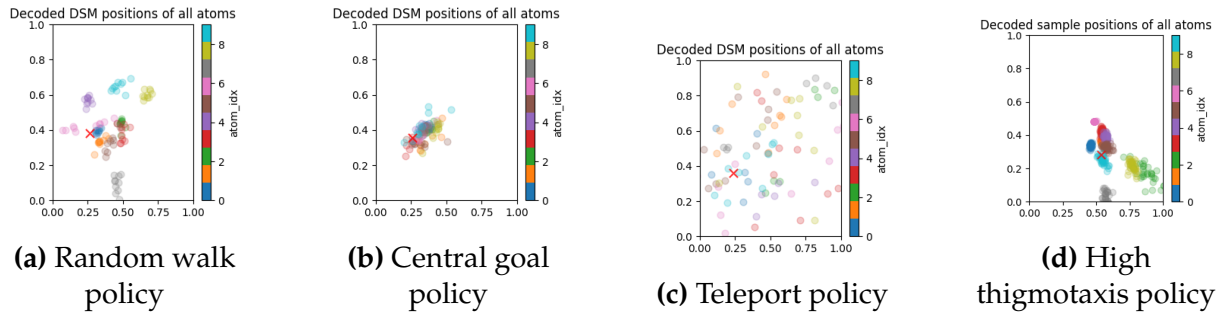


Figure 4.19: Predicted successor samples for a single source state - each subfigure represents a model trained on different policies

We investigate the model's ability to learn the representations of each policy by visualizing each atom's averaged rate maps. This highlights the diversity in representations among the atoms, with each potentially capturing different aspects of the environment's predictive structure. Figure 4.20 presents the results of averaging the DSM's PC rate maps in the random walk datasets, Figure 4.21 presents the same for the high Thigmotaxis dataset, Figure 4.22 for the central goal dataset, Figure 4.23 for the teleport dataset, Figure 4.24 for the wall env dataset.

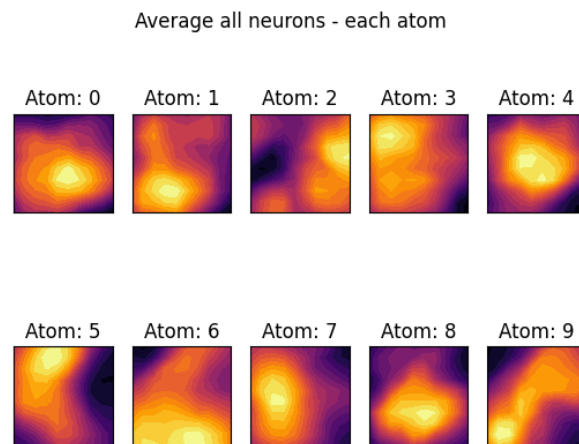


Figure 4.20: DSM Model trained on Random Walk policy: Averaged rate maps across all PCs for each atom, illustrating diversity in representations.

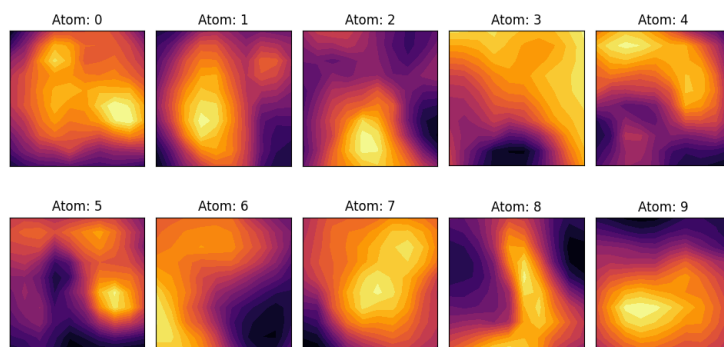


Figure 4.21: DSM Model trained on high Thigmotaxis policy: Averaged rate maps across all PCs for each atom, illustrating diversity in representations.

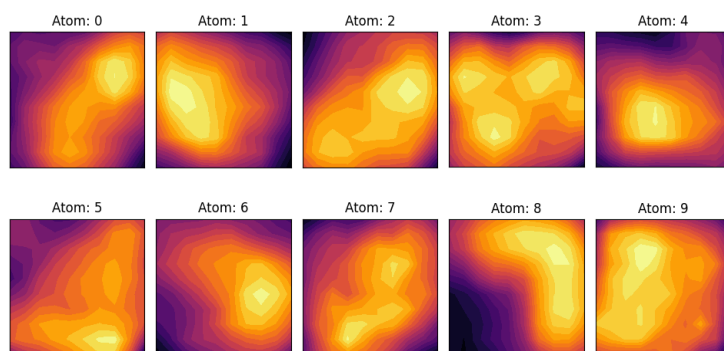


Figure 4.22: DSM Model trained on goal policy: Averaged rate maps across all PCs for each atom, illustrating diversity in representations.

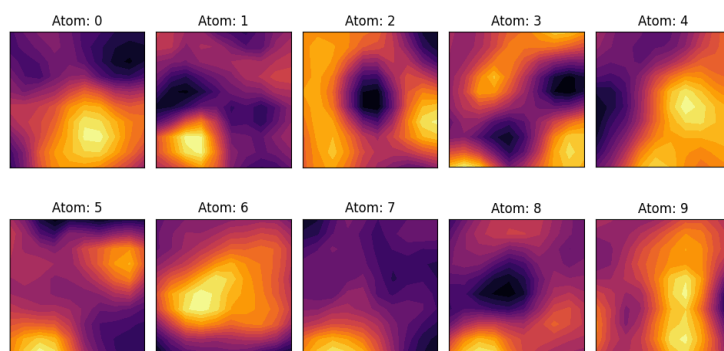


Figure 4.23: DSM Model trained on teleport policy: Averaged rate maps across all PCs for each atom, illustrating diversity in representations.

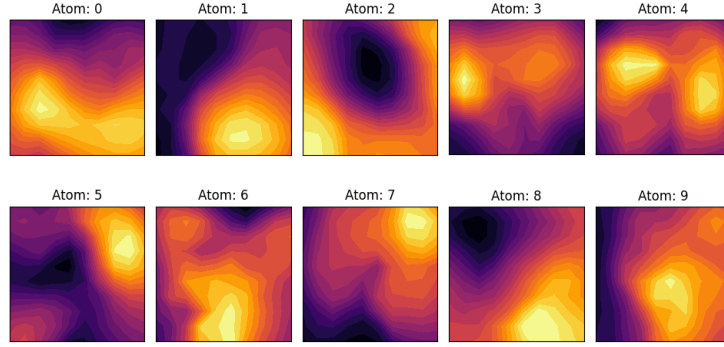


Figure 4.24: DSM Model trained on wall policy: Averaged rate maps across all PCs for each atom, illustrating diversity in representations.

The remaining analyses are shown on the model learned from the random walk dataset, as the other models exhibited similar patterns. The analyses for other policies have been shown in separate notebooks in the code repository for each model.

### t-SNE Visualization

We applied t-distributed Stochastic Neighbor Embedding (t-SNE) to visualize the DSM's internal representations for the random walk policy. Figure 4.25 shows the embeddings projected into two-dimensional space. The left panel displays the original states to be mapped, while the right panel shows the corresponding model embeddings after t-SNE projection.

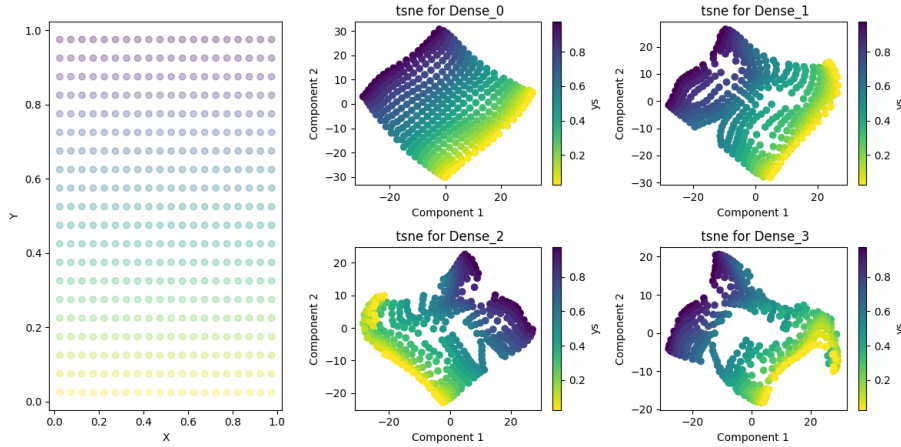


Figure 4.25: t-SNE visualization of DSM embeddings under the random walk policy. Left: Original states. Right: Model embeddings after t-SNE projection for each layer

## Linear Decoder Analysis

We employed a linear decoder to assess how well the DSM's internal activations can predict the source and predicted future states.

Figure 4.26 shows the decoding accuracy for the source states under the random walk policy, while Figure 4.27 illustrates the decoding accuracy for the predicted future states.

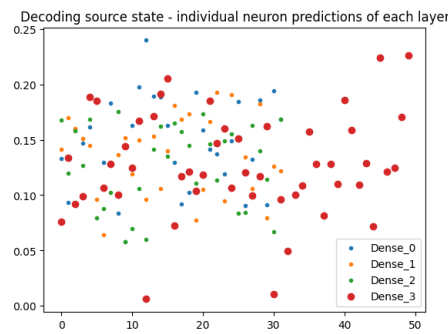


Figure 4.26: Decoding accuracy of source states from DSM activations under the random walk policy.

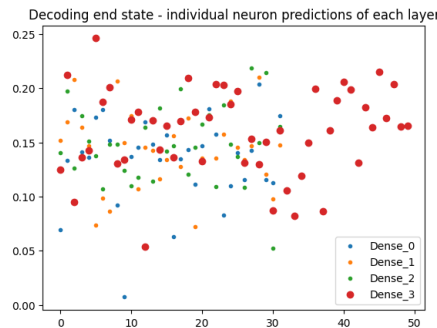


Figure 4.27: Decoding accuracy of predicted future states from DSM activations under the random walk policy.

## Entropy Analysis

We computed the entropy of the DSM's activations to understand how information is distributed across the network layers under the random walk policy. Figure 4.28 shows the entropy values for Atom 0.

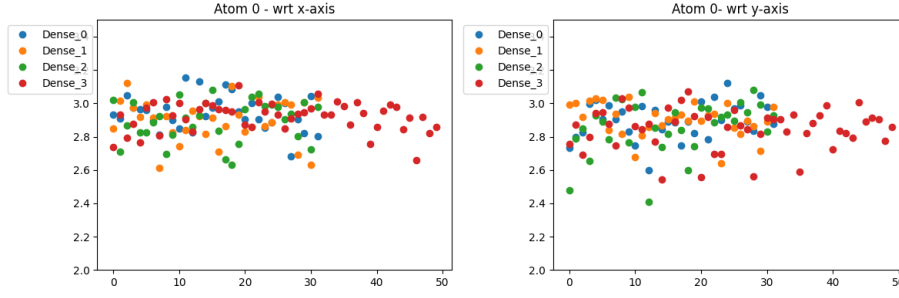


Figure 4.28: Entropy of DSM activations for Atom 0 under the random walk policy.

### Comparison Between SR and DSM

To understand systematic model biases in each atom of the DSM, we captured the activations of the intermediate layers and visualized the average rate maps for both the Successor Representation (SR) model and all atoms of the Distributional Successor Measure (DSM). By averaging across all place cells (PCs) as a population measure, we aimed to uncover any preferred trajectories or patterns that the DSM atoms might exhibit, given that each atom is supposed to model a different predictive map.

Figure 4.29 shows the averaged PC rate map for the SR model, where we observe a central tendency in the rate maps. This is expected since the input place cells were all Gaussians randomly positioned in the environment, and the agent followed a random trajectory.

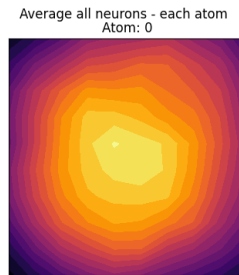


Figure 4.29: SR Model: Averaged rate map across all PCs, showing a central tendency.

In contrast, Figure 4.30 presents the results of averaging the DSM's PC rate maps separately for each atom. This highlights the diversity in representations among the atoms, with each potentially capturing different aspects of the environment's predictive structure.

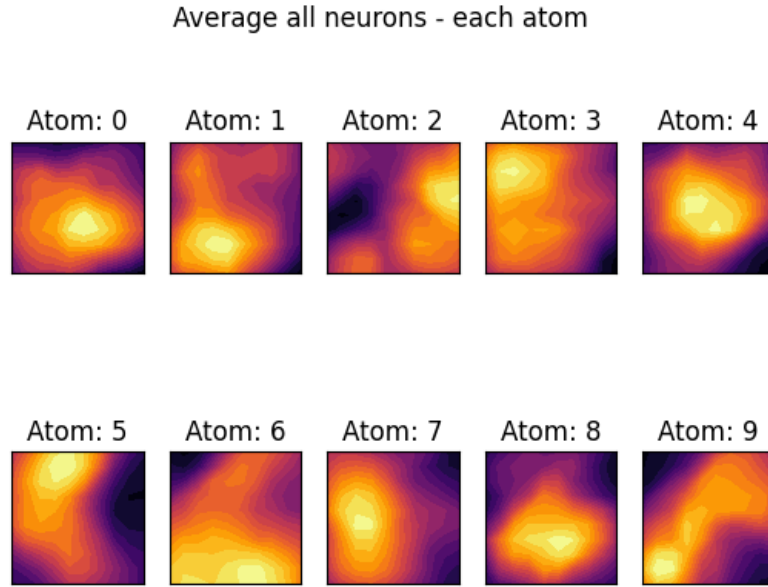


Figure 4.30: DSM Model trained on Random Walk: Averaged rate maps across all PCs for each atom

### Testing Generalization

To assess the DSM model's robustness to environmental changes, we tested its generalization by incrementally altering the positions of 1 to 50 place cell centers (PCs) within the environment. The modified firing activities of these PCs were then input into the model.

Figure 4.31 illustrates the averaged (across atoms) rate maps of place cell 13 in the DSM model as the number of altered PCs increases. Despite changes to up to 41 PCs, some of the DSM's rate maps remain relatively stable, demonstrating its ability to maintain consistent representations even as the environment changes.



Figure 4.31: DSM Model: Averaged rate maps of place cell 13 with increasing numbers of altered place cell centers ([1, 11, 21, 31, 41]).

In contrast, the SR model's rate maps for the same place cell become increasingly dis-

torted as more PCs are altered, indicating a lower tolerance to environmental changes (Figure 4.32).

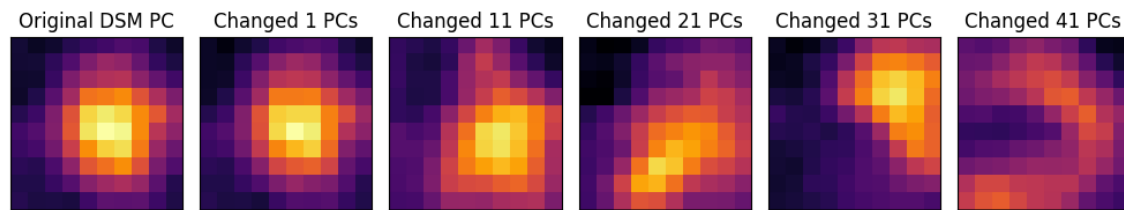


Figure 4.32: SR model: Averaged rate map of place cell 13 with increasing numbers of altered place cell centers

To quantify the models' robustness, we computed the cosine similarity between the original and modified predicted rate maps for both models. Figure 4.33 shows that the DSM model maintains comparatively higher similarity scores across all levels of alteration, while the SR model's similarity decreases more rapidly as the number of altered PCs increases. 4.34 shows the scores averaged across all the PCs. This indicates that the DSM is more robust to changes in the environment compared to the SR model. Figure

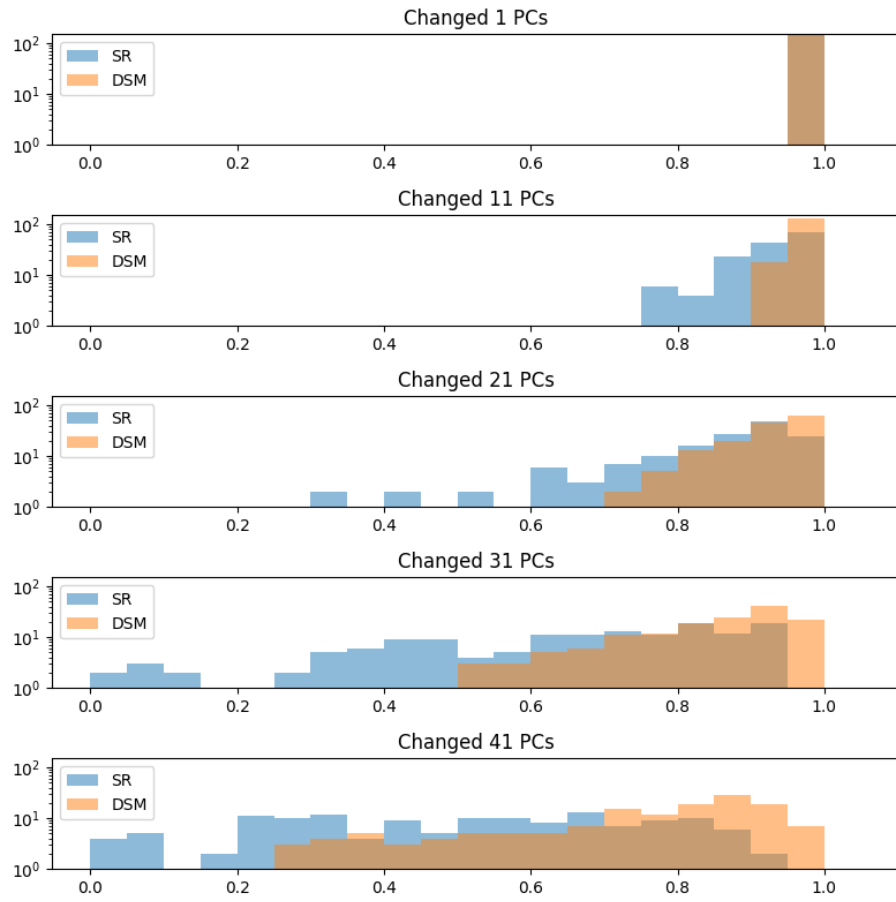


Figure 4.33: Cosine similarity between the original and modified predicted rate maps for the DSM and SR models

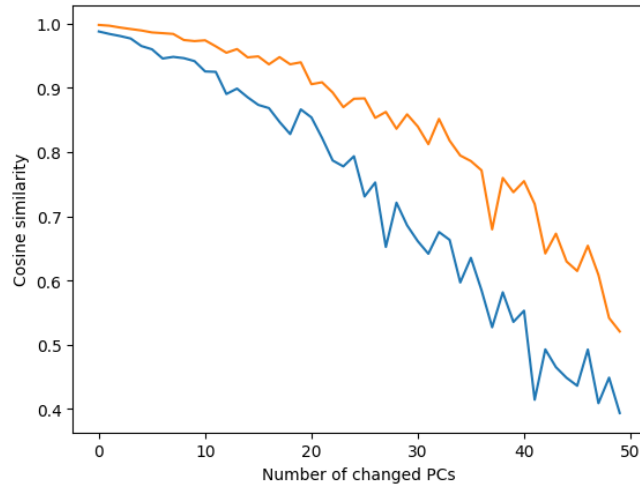


Figure 4.34: Cosine similarity between the original and modified predicted rate maps for the DSM and SR models, averaged over the samples

We observe that while both models are affected by changes in place cell remapping, the DSM is less susceptible to distortion than the SR model. This highlights that the DSM's ability to represent a distribution over future states makes it more resilient to environmental modifications compared to the SR's single expected future state representation.

## Chapter 5

# Discussion and Conclusion

Modeling the hippocampus is a complex and challenging task that requires striking a balance between abstracting away biological details while still capturing the diverse characteristics of place cells. In this study, we explored the features and capabilities of the Distributional Successor Measure (DSM) as a model that mirrors hippocampal functions in spatial navigation and memory

Our investigations involved analyzing the DSM’s internal representations through dimensionality reduction techniques, entropy calculations, and linear decoders in the Pendulum environment. We also examined its representations under different movement policies in place cell experiments. Additionally, we compared the DSM’s robustness to environmental changes with that of the traditional Successor Representation (SR) model, particularly in the context of hippocampal remapping phenomena.

Our findings indicate that the DSM captures the underlying structure of environmental dynamics to a certain degree. The t-SNE visualizations revealed that the DSM organizes states according to the topology induced by the environment’s dynamics, with similar environmental features arranged together. This supports the notion that the DSM partitions states based on environmental dynamics, capturing local and global structures necessary for accurate predictions and generalization.

In our place cell experiments, we model the inputs to the hippocampus with Gaussian place fields, which while a simplification, may serve as a reasonable abstraction based on evidence that spatially-tuned cells are found across various brain regions [28]. This abstraction allows us to focus on the core dynamics of spatial representation without the complexities of more detailed neural modeling.

Different movement policies, such as random walks, goal-directed behaviors, and thigmotaxis (boundary adherence), were shown to induce varying representations in the DSM. By comparing the spatial structure of the learned rate maps across different policies, we observed that the DSM adapts its internal representations to reflect the specific dynamics of each policy, while still modeling the environmental features. This adaptability suggests that the DSM can adjust its internal representations based on task-relevant information embedded in different movement strategies. However, this can be a slight disadvantage in situations where a consistent, unbiased representation of the environment is desired, as the DSM may encode movement-specific biases.

We also compared the DSM with the traditional SR model in representing spatial information. By visualizing the average rate maps for both models, we uncovered systematic biases or preferred trajectories exhibited by each atom of the DSM. While the SR model’s averaged place cell rate map showed an even distribution—consistent with the expectation due to random Gaussian input place cells and random agent trajectories—the DSM’s atoms exhibited diverse patterns in their averaged rate maps. This diversity indicates that each atom models a different predictive map, enabling the DSM to represent a distribution over future states. Consequently, the DSM provides a richer and more flexible representation compared to the SR model.

Assessing the DSM’s robustness to environmental changes was a critical aspect of our study. By incrementally altering the input activity at each source, we simulated changes analogous to modifying environmental cues or introducing obstacles. The DSM maintained higher cosine similarity scores between the original and modified predicted rate maps compared to the SR model. Visualizations of the averaged rate maps for a representative place cell showed that the DSM’s representations remained relatively stable even as the input activity drastically changed, whereas the SR’s rate maps became much more distorted under the same conditions. These findings highlight the DSM’s superior ability to adapt to environmental modifications, aligning with the hippocampus’s ability to maintain function despite alterations in the environment.

Our analysis demonstrates that the DSM outperforms the traditional SR model in some key aspects:

- **Rich Internal Representations:** Through dimensionality reduction, we observed that the DSM organizes internal representations to reflect environmental topology.

- **Diversity in Predictive Maps:** Each atom in the DSM models different predictive maps, providing a diverse set of representations that enhance the model’s ability to anticipate various potential futures.
- **Robustness to Environmental Changes:** The DSM shows greater resilience to environmental alterations, maintaining stable representations despite shifts in place cell centres.

These results underscore the DSM’s ability to represent a distribution over future states, making it more adaptable to environmental modifications. In contrast, the SR model, which relies on a single expected future state, is less robust to changes. The DSM’s resilience can be attributed to its architecture, where multiple atoms model different predictive maps, allowing it to capture a broader range of possible futures and providing redundancy against alterations in input.

Our findings align with biological observations in neuroscience. In the hippocampus, place cells exhibit heterogeneous responses to environmental changes, with some cells remapping while others remain stable [35]. The DSM mirrors this phenomenon by showing that while some of its internal representations are affected by changes, others maintain their predictive accuracy, collectively contributing to a stable overall representation. This suggests that the DSM provides a more biologically relevant model of hippocampal function compared to the traditional SR model.

### 5.0.1 Study Limitations

Although our results are encouraging, there are limitations to our study.

One limitation of my study is that the entropy and t-SNE analyses were conducted with respect to the source states (inputs) rather than the model’s predictions. In the pendulum experiments, Atoms 6 and 7 appear to predict specific ranges of angular velocities, indicating a sensitivity to particular dynamic behaviors, while Atom 1 primarily predicts stable equilibrium points. By focusing the analysis on the inputs instead of the predictions, potential insights into how the model encodes and differentiates outputs, particularly in terms of angular velocity and orientation, may have been overlooked. Future analysis should examine the entropy and t-SNE results based on the model’s predictions to better understand the atom-specific behaviors and output variability.

We have also not investigated the optimal hyperparameters of the DSM concerning the RatInABox environment. Conducting a systematic exploration of hyperparameters, such as the number of atoms, and network architecture, could improve its performance. Understanding how different architectural choices affect the model’s predictive accuracy and adaptability would be valuable for refining the DSM and enhancing its representations.

Our model assumes Gaussian-centered place fields as input, which, while a common simplification, may not capture the full complexity of inputs to biological place cells. Future work could involve incorporating sensory inputs directly to derive SRs from egocentric cues or integrating other known biological phenomena to enhance realism. Additionally, we used a fixed number of place fields, whereas in biological agents, the number and distribution of place fields can vary. Future studies could explore adaptive mechanisms for place field representation.

Moreover, a limitation of the DSM is that it assumes a fixed policy and transition dynamics, which restricts its usefulness in situations where policies change over time. While this is useful for robust policy evaluation, it is not suitable for transferring knowledge to arbitrary downstream tasks [77], an ability that is essential for cognitive flexibility in biological systems.

## 5.1 Future Work

Several avenues for future research could further enhance the DSM’s capabilities and biological plausibility:

**Improving Disentanglement of Representations:** Implementing techniques to encourage disentanglement in the DSM’s internal representations—such as constraining activations to be positive or introducing energy constraints—could enhance its ability to capture distinct task or environmental features [78]. These modifications may lead to more efficient and interpretable representations. Moreover, the addition of energy constraints and sparse connectivity, as observed in biological neural networks, could make the DSM more biologically plausible by better reflecting the optimization processes occurring in the brain [79].

**Incorporating Recurrent Connections:** Adding recurrence to the DSM could improve its ability to handle tasks involving memory and sequential processing, such as hippocampal replay and serial learning [80]. This addition would make the model more biologically plausible, as recurrent connections are prevalent in the hippocampus [81] and are essential for representing temporal sequences.

**Integrating Additional Neural Representations:** Incorporating representations like grid cells, head direction cells, or boundary vector cells might enhance the DSM’s navigational capabilities and biological plausibility. This integration could provide a more comprehensive model of spatial navigation, aligning with the multi-component nature of the hippocampal formation [24].

**Behavioral Feature Representation in Latent Variables:** Experiments could focus on training the DSM on multiple movement policies as different episodes in the same dataset. The investigation could explore whether DSM’s atoms capture specific behavioral features in a disentangled manner, such that one atom captures boundary adherence, while another captures goal-directed movement. This could also help understand if the model can capture multiple representations of the same environment, as evidenced in the hippocampus [82].

**Testing Predictions in Animal Models:** Reciprocal experiments could be conducted by testing the DSM model’s predictions in animal models, particularly in rodents using virtual reality (VR) environments. Cues can be manipulated to assess how changes in environmental inputs affect the model’s internal representations. This could involve altering boundary cues or introducing novel environmental features to investigate how the DSM adjusts its successor measures and behavioral predictions.

By pursuing these directions, future research can build upon our study, further advancing the understanding of predictive representations in artificial intelligence and their connections to biological systems. Such efforts could lead to more sophisticated models of spatial cognition, and have applications in neuroscience research and robotics.

## 5.2 Conclusion

In this study, we explored the capabilities of the Distributional Successor Measure model in modeling predictive representations for spatial navigation and control tasks. Motivated by the need for models that can robustly handle uncertainty and adapt to dynamic environments—paralleling the flexibility observed in biological neural systems—we explored the DSM model in terms of internal representations, adaptability and resilience to environmental changes.

Using environments like the inverted pendulum task and a simulated rat-foraging scenario, we employed techniques such as dimensionality reduction, entropy analysis, and linear decoders to delve into the DSM’s internal mechanisms. We also simulated environmental alterations to test the DSM’s robustness and compared its performance with that of the SR model.

Our findings address our research aims by demonstrating that the DSM can effectively model a rich set of predictive representations and maintain adaptability in dynamic environments. It outperforms the traditional SR model in capturing environmental dynamics and offers a framework that aligns more closely with biological neural processing.

Our results suggest that the DSM not only aligns more closely with biological observations of hippocampal function but also holds promise for practical applications requiring robust spatial representations. The DSM’s ability to maintain stable representations despite environmental alterations indicates its potential for use in real-world environments, enhancing the performance and reliability of systems that rely on spatial cognition and navigation. By bridging insights from neuroscience and artificial intelligence, this investigation offers a pathway toward more sophisticated models of spatial cognition.

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